

A Dual-Layer Cell-Partition Control System for Bioreactors: Provably-Stable Regulation with an Additive Confirmation Layer

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Abstract

We present a control system for stirred-tank bioreactors built on a strict architectural separation between two layers that most designs conflate. The *control layer* normalises every heterogeneous process channel—dissolved oxygen, pH, temperature, agitation, feed rate—onto a common bounded coordinate hypercube, partitions that hypercube into a finite set of *cells*, and dispatches a pre-compiled action from the cell that the current normalised state occupies. We prove that this quantised controller is globally asymptotically stable under an explicit piecewise Lyapunov condition, that its discrete-time operator is a contraction with geometric convergence, that its cell width is bounded above by a sampling (Nyquist-type) criterion tied to the process bandwidth, and that it is *structurally incorruptible*: because the controller consumes only a scalar timing/registry index and dispatches from a fixed finite action set, no sensor payload can drive the plant outside that set. The *monitoring layer* sits strictly downstream of the control layer across a one-directional interface. It treats each measurement not as a number but as one *confirmation* in a redundant ensemble: sensors are modelled as coupled oscillators that federate by phase entrainment, and a process excursion is confirmed only when independent channels agree, with a decline-to-safe-state outcome when they irreducibly disagree. Because observation and computation are the same operation—both place a value into a cell—the system runs two concurrent scripts, a sensor script and a *metabolic* script, the latter synthesising *virtual sensors*: channels that no probe measures but that are computed at a node from several physical chan-

nels (for example a respiratory-state channel assembled from pH, off-gas CO₂, and temperature) and then serve as genuine control coordinates, made safe-by-construction by a certified fallback value. We derive the *sensor synthesis map* that produces such channels: an information-preserving bijection switches each measurement between a time-domain value and a frequency-domain oscillator phase, inputs are combined in the frequency domain under a mean-recovery (circular-mean) constraint, and the result is switched back to a time-domain cell—one operation yielding both an actionable synthesised value and, as the concentration of the combination, a trustworthy ensemble invariant. Among computed channels we distinguish *ensemble invariants*: quantities invariant under relabelling of interchangeable units and realised by no single unit—information true of the whole sensor set and present in none of its members—which we prove are robust to any single-unit fault by an $O(1/n)$ bound and which detect exactly the correlated failures that redundant probes cannot. We show that the monitoring layer is *additive*: every one of its guarantees can fail—sensors drop out, the ensemble decoheres, confirmation goes silent—without affecting the control layer’s stability or safety proofs, because the interface is one-directional and the control law never depends on any monitoring output for correctness. In the worst case the system degrades continuously to an ordinary provably-stable single-channel cell-partition controller. We give the complete state-space model, stability proofs, a CUSUM-equivalent excursion-detection identity, a federation admission criterion, six algorithms, and a validation protocol on a fed-batch *Saccharomyces cerevisiae* process. The framework is self-contained:

every claim is proved from stated definitions using only standard results in nonlinear control, information theory, and synchronisation.

Keywords: bioreactor control; cell-partition quantisation; piecewise Lyapunov stability; S -normalisation; dissolved-oxygen control; CUSUM; sensor federation; Kuramoto synchronisation; virtual soft sensors; structural incorruptibility; fail-safe degradation.

1 Introduction

1.1 The problem

A stirred-tank bioreactor is a small chemical plant whose product is living cells or the molecules they secrete. Its controller must regulate a handful of coupled variables—dissolved oxygen (DO), pH, temperature, agitation speed, and one or more feed rates—each measured in incommensurable units and each coupled to the others through the metabolism of the culture (Doran, 2013; Shuler et al., 2017). Three difficulties recur across the bioprocess-control literature (Alford, 2006; Mandenius, 2016). First, *dimensional heterogeneity*: composing a DO channel (percent of saturation) with a temperature channel (kelvin) and a feed channel (litres per hour) requires unit bookkeeping at every algebraic step, and errors in that bookkeeping are a documented source of instability. Second, *sensor unreliability*: the DO and pH probes that dominate bioprocess feedback drift, foul, and fail, and a controller that trusts a single probe inherits that probe’s failure modes (Garcia-Ochoa and Gomez, 2009; Luttmann et al., 2012). Third, *safety and integrity*: a networked controller for a pharmaceutical-grade process is a cyber-physical system whose failure or compromise is costly, and the regulatory framework demands that process excursions be detected and that the system fail to a defined safe state (Cárdenas et al., 2008; U.S. Food and Drug Administration, 2004).

The three difficulties pull in different directions. Richer sensing improves observability but enlarges the attack and failure surface. Tighter multivariable control improves regulation but couples the controller’s correctness to the reliability of every channel it reads. A design that fuses all sensors into one state estimate and feeds that estimate to one control law is only as trustworthy as its least reliable sensor.

1.2 The thesis: separate the layers

This paper argues that the correct response is an architectural separation, enforced by a one-directional interface, between a *control layer* and a *monitoring layer*.

The control layer is deliberately austere. It maps every physical channel onto a common bounded coordinate—we call this *S-normalisation*—so the working state lives in a dimensionless hypercube $\mathcal{H} = [0, 1]^m$. It partitions $\mathcal{H}$ into a finite set of cells and stores, for each cell, one pre-compiled control action. At each step it determines which cell the current normalised state occupies and dispatches that cell’s action. Nothing else. This austerity is what makes the layer provable: we establish global asymptotic stability by a piecewise Lyapunov argument, geometric convergence by a contraction argument, a cell-width bound by a sampling criterion, and—because the dispatch depends only on cell membership and the action set is fixed at compile time—structural incorruptibility, meaning no measurement value can cause an action outside the pre-approved set.

The monitoring layer is where richness lives, and it is placed entirely downstream. It reads the same physical channels, but instead of producing a state estimate that the controller must trust, it produces *confirmations*: each measurement is one vote in a redundant ensemble that agrees or dissents about whether the process has entered a given region. Sensors federate—each is modelled as an oscillator whose frequency is perturbed by what it measures, and the ensemble reaches agreement by phase entrainment rather than by exchanging content. When the ensemble is coherent, agreement is automatic and cheap; when it is not, the layer degrades gracefully and, in the limit of disagreement, declines to a safe state. Crucially, the monitoring layer’s outputs enter the control layer only through the same narrow interface any operator would use—as a request to change set-point or enter a safe mode—never as a term in the control law. It is therefore *additive*: it can only improve confirmation quality and diagnostic resolution, and it cannot break the control-layer proofs, because those proofs do not reference it.

1.3 The worst-case guarantee

The value of the separation is a precise worst-case statement, which we prove in Section 9: *if every component of the monitoring layer fails, the system reduces to an ordinary single-channel cell-partition controller, which remains globally asymptotically*

stable and structurally incorruptible. There is no scenario in which the speculative or elaborate parts of the monitoring layer are load-bearing for stability or safety. This is what licenses building the monitoring layer ambitiously: its failure is bounded by the control layer’s floor.

1.4 Contributions

1. **S-normalisation and the cell-partition control law** (Sections 2 and 3). A dimensionless bounded state space $\mathcal{H} = [0, 1]^m$, a finite cell partition, and a nearest-centroid proportional dispatch that is L^2 -optimal among piecewise-constant laws.
2. **Stability and convergence** (Section 4). Piecewise Lyapunov global asymptotic stability reducible to per-cell linear matrix inequalities; a Banach contraction with explicit geometric rate; and a Nyquist-type cell-width bound tied to process bandwidth.
3. **Structural incorruptibility** (Section 5). A proof that no measurement payload has a path to an action outside the pre-compiled set, and a bounded-effect analysis of spurious or delayed sensor events.
4. **The confirmation calculus** (Section 6). A CUSUM-equivalent identity linking cell-boundary crossings to out-of-control alarms; a multiplicative composition law for independent sensor confirmations; a coherence requirement forcing at least three independent channels for robust grounding; and a convergent/decline dichotomy for honest termination.
5. **Sensor federation** (Section 7). A temporal-agent model in which measurements join an ensemble by oscillator perturbation, a critical-coupling threshold for phase lock, a zero-overhead theorem at coherence, and an information-theoretic admission criterion.
6. **Virtual sensors, ensemble invariants, and the two-script model** (Section 8). Using the equivalence of observation and computation, a concurrent *metabolic script* that synthesises channels no probe measures—local virtual sensors (functions of named channels, used to act) and *ensemble invariants* (relabelling-invariant, unit-less quantities true of the whole set and present in none, used to trust)—which serve as genuine control coordinates, kept safe-by-construction by a fallback guard and certified

on the extended grid, with an $O(1/n)$ single-unit-fault robustness bound for invariants.

7. **The sensor synthesis map and the time–frequency switch** (Sections 2.4 and 8.7). A derived, information-preserving bijection between a channel’s time-domain value and its frequency-domain phase that respects the cell partition, and a synthesis operation (frequency-domain circular mean under mean-recovery, switched back to a cell) that generates the value and the invariant of a computed channel together.
8. **The separation theorem and fail-safe degradation** (Section 9). A formal proof that the monitoring layer is additive and that worst-case behaviour is an ordinary stable controller.
9. **Bioreactor realisation** (Sections 10 and 11). The full architecture, channel definitions, six algorithms, and a validation protocol on a fed-batch yeast process.

1.5 Method and scope

Every substantive statement is a definition, a theorem proved here, or a citation of a standard result compared against such a theorem. The mathematics is drawn from nonlinear and quantised control (Brockett and Liberzon, 2000; Khalil, 2002), functional analysis (Kreyszig, 1989), statistical process control (Montgomery, 2020; Page, 1954), information theory (Cover and Thomas, 2006; Shannon, 1948), and synchronisation theory (Kuramoto, 1984; Strogatz, 2000). Where a claim in the monitoring layer rests on an assumption we cannot discharge in full generality, we mark it explicitly and confine it to the additive layer, so that no control-layer guarantee inherits it. The paper is self-contained: a reader needs no prior familiarity with any companion work to follow every proof.

1.6 Organisation

Section 2 fixes the bounded state space, S-normalisation, the cell partition, the resolution floor, and the time–frequency switch between value and phase. Section 3 defines the control law. Section 4 proves stability, convergence, and the sampling bound. Section 5 proves structural incorruptibility. Section 6 develops the confirmation calculus. Section 7 develops sensor federation. Section 8 develops virtual sensors and the two-script model. Section 9 proves the separation theorem and characterises worst-case degradation. Section 10 instan-

tiates the framework for a stirred-tank bioreactor. Section 11 gives the algorithms. Section 12 gives the validation protocol. Section 13 discusses scope and limitations. Section 14 concludes.

2 Foundations: bounded state, normalisation, and cells

The control layer operates on a normalised, bounded, finitely-partitioned state space. This section constructs that space, records the elementary facts we will use, and derives the resolution floor—the strictly positive lower bound on how finely any physical channel can be resolved, which every later bound refers back to.

2.1 Physical channels and S-normalisation

Definition 2.1 (Process channel). A *process channel* is a measured or manipulated physical quantity $q \in \mathbb{R}$ of the bioreactor with a known operating range $[q_{\min}, q_{\max}]$, $q_{\min} < q_{\max}$. For a stirred-tank reactor the channels are dissolved oxygen, pH, temperature, agitation rate, and one or more feed rates (Section 10).

Definition 2.2 (S-coordinate). The *S-coordinate* of a channel value q is the affine normalisation onto the unit interval,

$$S(q) = \frac{q - q_{\min}}{q_{\max} - q_{\min}} \in [0, 1], \quad (1)$$

with inverse $q = q_{\min} + S(q)(q_{\max} - q_{\min})$. For an m -channel reactor the *S-state vector* is $\mathbf{S} = (S(q_1), \dots, S(q_m))^T \in [0, 1]^m$.

The single purpose of (1) is to remove dimension. After normalisation every channel lives in the same dimensionless interval, so channels of incommensurable physical units can be compared, composed, and partitioned uniformly, with no conversion factors. This is the resolution of the dimensional-heterogeneity difficulty of Section 1.

Definition 2.3 (S-state space). The *S-state space* is the hypercube $\mathcal{H} = [0, 1]^m \subset \mathbb{R}^m$ with the Euclidean norm $\|\mathbf{S}\| = (\sum_{i=1}^m S_i^2)^{1/2}$ and the metric $d(\mathbf{S}, \mathbf{S}') = \|\mathbf{S} - \mathbf{S}'\|$.

Proposition 2.4 (Compact complete metric space). $(\mathcal{H}, d)$ is a compact, connected, complete metric space, and it carries the Lebesgue measure λ of $\mathbb{R}^m$ restricted to $\mathcal{H}$.

Proof. $\mathcal{H} = [0, 1]^m$ is a closed and bounded subset of $\mathbb{R}^m$, hence compact by Heine–Borel and complete as a closed subset of the complete space $(\mathbb{R}^m, \|\cdot\|)$ (Munkres, 2000). It is a product of connected intervals, hence connected. Restriction of Lebesgue measure to a Borel set is a measure. $\square$

Compactness and completeness are exactly what the extreme-value, finite-cover, and fixed-point arguments of Section 4 require; we invoke this proposition without further comment there.

2.2 The cell partition

Definition 2.5 (Cell partition). A *cell partition* of $\mathcal{H}$ is a finite family $\mathcal{C} = \{C_1, \dots, C_N\}$ of Borel sets with (i) $\bigcup_{i=1}^N C_i = \mathcal{H}$; (ii) $C_i \cap C_j = \emptyset$ for $i \neq j$; and (iii) $\lambda(C_i) > 0$ for every i . Each C_i is a *cell*. For axis-aligned rectangular cells, the *width* of the partition is the smallest cell side length along any coordinate,

$$w_{\text{cell}} = \min_i \min_{1 \leq k \leq m} \text{len}_k(C_i), \quad (2)$$

where $\text{len}_k(C_i)$ is the extent of cell C_i along axis k . A partition is *w_{cell}-regular* if every cell contains a coordinate-aligned box of side $\geq w_{\text{cell}}$ in each dimension.

For the reactor we use axis-aligned rectangular cells (a grid, possibly non-uniform), for which the width is simply the smallest grid spacing along any axis. Rectangular cells make cell lookup an $O(m)$ operation (one comparison per axis) and make the per-cell stability tests of Section 4 tractable.

Definition 2.6 (Cell of a state; cell map). For $\mathbf{S} \in \mathcal{H}$ let $\text{cell}(\mathbf{S})$ denote the unique cell containing $\mathbf{S}$ (ties on measure-zero boundaries broken by a fixed rule, e.g. half-open cells). The *cell map* $\text{cell}: \mathcal{H} \rightarrow \mathcal{C}$ is measurable.

Definition 2.7 (Cell centroid). The *centroid* of cell C_i is

$$\bar{\mathbf{S}}_i = \frac{1}{\lambda(C_i)} \int_{C_i} \mathbf{S} d\lambda \in \overline{C_i}. \quad (3)$$

2.3 The resolution floor

No physical instrument resolves a channel to a point. Every sensor and every actuator has a finite resolution $\delta > 0$ below which two values are indistinguishable, set by quantisation, noise, or the uncertainty of the transduction. After S-normalisation this resolution becomes a dimensionless floor on cell width.

Assumption 2.8 (Finite resolution). Each channel k has a finite instrument resolution $\delta_k > 0$: values of q_k differing by less than $\delta_k(q_{k,\max} - q_{k,\min})$ in raw units are not distinguishable by the reactor’s transducers. Consequently no realisable cell partition has width below

$$\beta := \min_{1 \leq k \leq m} \delta_k > 0. \quad (4)$$

The constant β is the *resolution floor*.

Proposition 2.9 (The floor is strictly positive and bounds every cell). *Under Theorem 2.8, every w_{cell} -regular realisable partition has $w_{\text{cell}} \geq \beta > 0$, and every cell has Lebesgue measure $\lambda(C_i) \geq \beta^m > 0$.*

Proof. A realisable partition cannot separate two states that the instruments cannot distinguish; hence no cell side can be shorter than the smallest distinguishable increment β of (4), giving $w_{\text{cell}} \geq \beta$. A w_{cell} -regular cell contains a coordinate-aligned box of side $\geq w_{\text{cell}} \geq \beta$ in each of m dimensions, whose measure is $\geq \beta^m$. $\square$ $\square$

Remark 2.10 (Why the floor matters). The floor is the single quantity that makes the control layer finite and therefore provable. Because $\lambda(C_i) \geq \beta^m$ and $\lambda(\mathcal{H}) = 1$, the number of cells is bounded, $N \leq \beta^{-m} < \infty$ (Theorem 2.11), so the action set is finite; because $w_{\text{cell}} \geq \beta$, the sampling bound of Theorem 4.5 is non-vacuous; and because no separation is finer than β , a state’s cell membership is a robust, noise-tolerant datum rather than a fragile point estimate. Every guarantee in Sections 4 and 5 ultimately rests on $\beta > 0$.

Lemma 2.11 (Finite cell count). *Under Theorem 2.8, any realisable partition has $N \leq \lambda(\mathcal{H})/\beta^m = \beta^{-m} < \infty$.*

Proof. The cells are disjoint with $\sum_i \lambda(C_i) = \lambda(\mathcal{H}) = 1$ and each $\lambda(C_i) \geq \beta^m$ by Theorem 2.9; hence $N\beta^m \leq 1$. $\square$ $\square$

2.4 Two views of a normalised state: the time–frequency switch

The same S-state admits two descriptions, one used by each layer, and the map between them—the *sensor synthesis map*—is the operational form of the identity “observation = computation” developed in Section 8. The control layer names a state in the *time domain*: by its cell index i with $\mathbf{S} \in C_i$, reconstructed when needed to the centroid $\bar{\mathbf{S}}_i$; equivalently, per channel, by the inter-pulse

interval ΔP of Section 7. The monitoring layer names a state in the *frequency domain*: by an oscillator phase. We derive the switch between the two views, prove it is an information-preserving bijection that respects the cell partition, and show it is the substrate on which sensor synthesis (Section 8.7 and algorithm 7) operates.

Definition 2.12 (Phase representation). The *phase* of a normalised channel value $S \in [0, 1]$ is

$$\Phi(S) = 2\pi S \in [0, 2\pi), \quad (5)$$

the angular position of a reference oscillator whose one full turn spans the channel’s operating range. The value S is the *time-domain* coordinate (a position in $[0, 1]$, equivalently a cell index and offset); $\phi = \Phi(S)$ is the *frequency-domain* coordinate (an oscillator phase).

Proposition 2.13 (The switch is an information-preserving bijection). *The map $\Phi: [0, 1] \rightarrow [0, 2\pi)$ of (5) is an order-preserving bijection with inverse $\Phi^{-1}(\phi) = \phi/2\pi$; both Φ and Φ^{-1} are Lipschitz (constants 2π and $1/2\pi$) and measurable. Consequently the time-domain and frequency-domain descriptions of a channel carry identical information, and the switch loses none.*

Proof. $\Phi(S) = 2\pi S$ is affine with positive slope, hence a strictly increasing continuous bijection of $[0, 1]$ onto $[0, 2\pi)$ with continuous inverse $\phi/2\pi$; affine maps are Lipschitz with the stated constants and Borel measurable. A bijection with measurable inverse transports the Borel σ -algebra of one interval onto the other, so no measurable event is created or destroyed: the two coordinates encode the same information. $\square$ $\square$

Proposition 2.14 (The switch respects the cell partition). *The image under Φ of a one-dimensional cell $[a, b] \subseteq [0, 1]$ is the phase arc $[2\pi a, 2\pi b)$, and Φ carries the channel’s cell partition onto a partition of the phase circle into arcs of equal count. Hence cell membership is the same predicate in either domain: a time-domain cell test and a frequency-domain arc test return the same cell.*

Proof. Φ is an increasing bijection, so it maps the interval $[a, b]$ onto $[2\pi a, 2\pi b)$ and maps a partition of $[0, 1]$ into intervals onto the partition of $[0, 2\pi)$ into the corresponding arcs, preserving order and disjointness. Membership $S \in [a, b] \iff \phi \in [2\pi a, 2\pi b)$ by monotonicity. $\square$ $\square$

Corollary 2.15 (Consistency with the frequency-perturbation form). *The frequency-perturbation encoding of a measurement used by the federation, $\Delta P = 2\pi/(\omega + \delta\omega)$ (Section 7), is the time-domain image of the phase view: a measured shift $\delta\omega$ of the oscillator frequency advances the phase, and reading the phase back through Φ^{-1} recovers the S -value and hence the cell. The control layer therefore never needs the phase view—it reads S directly—while the monitoring layer uses it only to federate sensors (Section 7).*

Proof. Over one reference period the accumulated phase is $\phi = \int (\omega + \delta\omega) dt$; normalising by 2π and applying Φ^{-1} returns the corresponding $S \in [0, 1]$, whose cell is the same in both domains by Theorem 2.14. The inter-pulse interval $\Delta P = 2\pi/(\omega + \delta\omega)$ is the time-domain reading of the same oscillator. $\square$ $\square$

Theorems 2.13 to 2.15 establish the switch as a lossless, cell-preserving bijection between a time-domain value and a frequency-domain phase. This is what lets the two layers use different views of one state without either losing information, and—crucially—it is the substrate for *synthesising* new channels by combining phases and switching back, which we derive in Section 8.7 once the virtual-channel machinery is in place. The clean split of views is also what the separation of Section 9 formalises: the control layer lives entirely in the time domain, the federation entirely in the frequency domain, and the synthesis map is the only bridge.

3 The cell-partition control law

We now define the control layer. It is a quantised feedback law in the sense of Brockett and Liberzon (Brockett and Liberzon, 2000): a finite-valued control that stabilises a continuous-state plant. The novelty for bioprocessing is that the partition is defined on the S -normalised hypercube, so a single law serves all channels uniformly, and that the design is arranged so the monitoring layer can be removed without disturbing it.

3.1 The plant model

We model the closed-loop reactor dynamics in the S -state space. Write $\mathbf{S}(t) \in \mathcal{H}$ for the normalised state and $\mathbf{u} \in \mathcal{U}$ for the applied control from a finite control set $\mathcal{U} = \{u_1, \dots, u_N\}$.

Assumption 3.1 (Cellwise-Lipschitz dynamics). The closed-loop S -dynamics are

$$\dot{\mathbf{S}} = \mathbf{F}(\mathbf{S}, \mathbf{u}), \quad (6)$$

where $\mathbf{F}: \mathcal{H} \times \mathcal{U} \rightarrow \mathbb{R}^m$ is Lipschitz in $\mathbf{S}$ on each cell C_i (with possibly cell-dependent Lipschitz constant L_i), and $\mathcal{H}$ is forward-invariant: trajectories starting in $\mathcal{H}$ remain in $\mathcal{H}$ (guaranteed physically by the saturation of every actuator at the channel limits $q_{\min}, q_{\max}$).

Forward invariance is not an idealisation: it holds because the normalisation (1) clamps each channel to $[0, 1]$ by construction, and the physical actuators saturate at the same limits.

3.2 The cell-action map and the control law

Definition 3.2 (Cell-action map). A *cell-action map* is a function $A: \mathcal{C} \rightarrow \mathcal{U}$ assigning to each cell one pre-compiled control action. The induced state-feedback law is

$$\mathbf{u}(t) = A(\text{cell}(\mathbf{S}(t))). \quad (7)$$

The controller’s entire runtime behaviour is (7): measure $\mathbf{S}$, find its cell, emit that cell’s stored action. The action is pre-compiled and pre-verified offline; at runtime no arithmetic on the raw measurement, no parsing of any payload, and no search occurs. This austerity is the source of both the incorruptibility of Section 5 and the additivity of Section 9.

Definition 3.3 (Nearest-centroid proportional map). Given a setpoint $\mathbf{S}^* \in \mathcal{H}$ and a scalar gain $K > 0$, the *nearest-centroid proportional* cell-action map is

$$u_i = K(\mathbf{S}^* - \bar{\mathbf{S}}_i), \quad i = 1, \dots, N, \quad (8)$$

i.e. each cell drives the state from its centroid toward the setpoint.

Proposition 3.4 (L^2 -optimality of the nearest-centroid map). *Among all piecewise-constant maps $A: \mathcal{C} \rightarrow \mathbb{R}^m$ that are constant on each cell, the nearest-centroid map (8) minimises the $L^2(\mathcal{H}, \lambda)$ deviation from the ideal proportional law $K(\mathbf{S}^* - \mathbf{S})$:*

$$A^* = \arg \min_A \int_{\mathcal{H}} \|K(\mathbf{S}^* - \mathbf{S}) - A(\text{cell}(\mathbf{S}))\|^2 d\lambda. \quad (9)$$

Proof. The objective separates over cells, since the cells partition $\mathcal{H}$. On a fixed cell C_i the value u_i minimising $\int_{C_i} \|K(\mathbf{S}^* - \mathbf{S}) - u_i\|^2 d\lambda$ is found by setting the derivative in u_i to zero: $u_i = \frac{1}{\lambda(C_i)} \int_{C_i} K(\mathbf{S}^* - \mathbf{S}) d\lambda = K(\mathbf{S}^* - \bar{\mathbf{S}}_i)$, the centroid rule (8). Minimising each cell’s integral minimises the sum. $\square$ $\square$

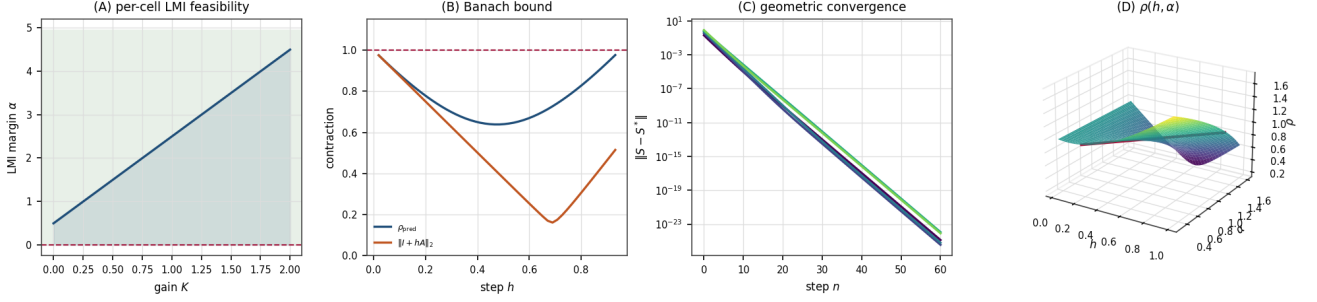


Figure 1: **Control-Layer Core: The Loop Is Provably Stable Before It Is Run.** (A) Per-cell LMI margin α on the linear y -axis (0 to ~ 4.5) versus proportional gain K on the linear x -axis (0 to 2), one point per gain over a 60-point sweep (steel-blue), with the feasibility floor $\alpha = 0$ marked (coral dashed) and the certified region shaded. The margin $\alpha = -\lambda_{\max}(\mathbf{A}_i + \mathbf{A}_i^\top)$ rises linearly and strictly above zero once the gain dominates the plant’s local expansion ($\alpha = 2.49$ at $K = 1$), the empirical content of Theorem 4.2: global asymptotic stability reduces to a finite family of per-cell linear matrix inequalities, all feasible here, while a deliberately under-damped configuration returns $\alpha = -1.15$ and is correctly rejected. (B) Contraction factor on the linear y -axis (0 to 1.15) versus integrator step h on the linear x -axis (0.02 to ~ 0.93), showing the Banach bound ρ_{pred} (steel-blue) and the exact one-step operator norm $\|I + h\mathbf{A}\|_2$ (coral), with the instability line $\rho = 1$ marked (crimson dashed). The predicted bound dominates the exact contraction at every step ($\rho_{\text{pred}} = 0.638$ against $\rho_{\text{exact}} = 0.408$ at the working step), the direct statement of Theorem 4.3: the discrete-time operator is a contraction with geometric rate $\rho < 1$ whenever $h < 2\alpha/L_F^2$, here with $\alpha = 1.25 \leq L_F = 1.62$. (C) Tracking error $\|\mathbf{S} - \mathbf{S}^*\|$ on a logarithmic y -axis (10^{-23} to 10^1) versus control step n on the linear x -axis (0 to 60), for five initial conditions (viridis colourmap). Every trajectory descends as a straight line on the semi-log axis: convergence is geometric at the certified rate, $\|T^n(\mathbf{S}_0) - \mathbf{S}^*\| \leq \rho^n \|\mathbf{S}_0 - \mathbf{S}^*\|$ (Equation (13)), reaching machine zero within sixty steps regardless of start. (D) Three-dimensional surface of the contraction factor $\rho(h, \alpha) = \sqrt{1 - 2h\alpha + h^2L_F^2}$ over integrator step h (0.02 to 1) and descent rate α (0.3 to 1.6), viridis colourmap, with the stability edge $h = 2\alpha/L_F^2$ traced (crimson), viewing angle elevation 22° azimuth -58° . The surface dips below $\rho = 1$ throughout the certified region and the crimson edge marks the Courant–Friedrichs–Lewy-type limit of Equation (12): the same grid-and-step constraint an explicit finite-volume solver must satisfy (Theorem 4.7).

3.3 Discrete-time realisation

A digital controller applies (7) at sampling instants. Let $h > 0$ be the sampling period and define the discrete-time operator

$$T(\mathbf{S}) = \mathbf{S} + h \mathbf{F}(\mathbf{S}, A(\text{cell}(\mathbf{S}))), \quad (10)$$

the forward-Euler step of (6) under the cell-partition law. The closed-loop trajectory is the orbit $\mathbf{S}_{k+1} = T(\mathbf{S}_k)$. The sampling period h is tied to the reference-oscillator/pulse rate $f_{\text{ref}} = 1/h$ used by the monitoring layer (Section 7), but the control law's correctness depends only on the bound of Theorem 4.5 and Theorem 4.3, never on the monitoring layer.

3.4 The committed-step record

The controller maintains a single monotone counter, used later for excursion accounting and replay resistance.

Definition 3.5 (Committed record). The *committed record* $M \in \mathbb{N}$ is the number of control steps taken since initialisation. It increases by one at each application of (10) and is never decremented.

Proposition 3.6 (Monotonicity and non-return). *M is strictly increasing in step index. Consequently the controller never returns to a prior state $(\mathbf{S}, M)$ even when it revisits a prior $\mathbf{S}$ -value: for step indices $j > i$, $(\mathbf{S}_j, M_j) \neq (\mathbf{S}_i, M_i)$ because $M_j > M_i$, regardless of whether $\mathbf{S}_j = \mathbf{S}_i$. An operation intended to undo a prior step is itself a step and increments M .*

Proof. Each step increments M by exactly one (Theorem 3.5), so $M_j = M_i + (j - i) > M_i$ for $j > i$; the second components of the two states differ, so the states differ. An “undo” is an application of (10), hence a further increment. $\square$ $\square$

The record makes “the same reading again” distinguishable from “the original reading,” which is exactly what a replay defence and an excursion audit require (Section 5). It costs one integer and is never load-bearing for stability.

4 Stability, convergence, and the sampling bound

This section establishes the three quantitative guarantees of the control layer: global asymptotic stability of the setpoint, geometric convergence of the discrete-time loop, and an upper bound on cell

width required for faithful regulation. All three are proved from the definitions of Sections 2 and 3 using standard results, and all three refer back to the resolution floor $\beta > 0$.

4.1 Piecewise Lyapunov stability

Theorem 4.1 (Piecewise Lyapunov global asymptotic stability). *Suppose there is a continuous $V: \mathcal{H} \rightarrow \mathbb{R}_{\geq 0}$ and class $\mathcal{K}_\infty$ functions $\alpha_1, \alpha_2, \alpha_3$ such that for all $\mathbf{S} \in \mathcal{H}$,*

- (i) $\alpha_1(\|\mathbf{S} - \mathbf{S}^*\|) \leq V(\mathbf{S}) \leq \alpha_2(\|\mathbf{S} - \mathbf{S}^*\|);$
- (ii) *on every cell C_i , along (6) with $\mathbf{u} = A(C_i)$,*
 $\dot{V}(\mathbf{S}) \leq -\alpha_3(\|\mathbf{S} - \mathbf{S}^*\|).$

Then the setpoint $\mathbf{S}^$ is globally asymptotically stable for the closed loop on $\mathcal{H}$.*

Proof. Condition (i) makes V positive definite and radially unbounded relative to $\mathbf{S}^*$ on $\mathcal{H}$ (Khalil, 2002). By (ii), on each cell V strictly decreases along trajectories; since the finitely many cells cover $\mathcal{H}$ (Theorem 2.11) and V is continuous across the measure-zero cell boundaries, the decrease holds along any trajectory, which crosses only finitely many cells in finite time. Integrating, $V(\mathbf{S}(t)) - V(\mathbf{S}(0)) = \int_0^t \dot{V} d\tau \leq -\int_0^t \alpha_3(\|\mathbf{S}(\tau) - \mathbf{S}^*\|) d\tau$. Because $\mathcal{H}$ is compact and forward-invariant (Theorem 3.1), the trajectory is defined for all $t \geq 0$ and remains in $\mathcal{H}$. Standard Lyapunov theory then gives $V(\mathbf{S}(t)) \rightarrow 0$ and, by the lower bound in (i), $\|\mathbf{S}(t) - \mathbf{S}^*\| \rightarrow 0$, which is global asymptotic stability (Khalil, 2002). $\square$ $\square$

Corollary 4.2 (Quadratic sufficiency: a per-cell LMI). *Take $V(\mathbf{S}) = \|\mathbf{S} - \mathbf{S}^*\|^2$. Then the conditions of Theorem 4.1 hold if, on each cell C_i with local linearisation $\mathbf{A}_i = \partial \mathbf{F} / \partial \mathbf{S}|_{C_i}$ (under action u_i), there is $\alpha_i > 0$ with*

$$\mathbf{A}_i + \mathbf{A}_i^\top \preceq -\alpha_i I. \quad (11)$$

Global asymptotic stability then reduces to checking the finite family of linear matrix inequalities (11), one per cell, solvable by standard semidefinite programming (Johansson and Rantzer, 1998).

Proof. With $V = \|\mathbf{S} - \mathbf{S}^*\|^2$, $\dot{V} = 2(\mathbf{S} - \mathbf{S}^*)^\top \mathbf{F}$. Linearising $\mathbf{F} \approx \mathbf{A}_i(\mathbf{S} - \mathbf{S}^*)$ on C_i gives $\dot{V} \approx (\mathbf{S} - \mathbf{S}^*)^\top (\mathbf{A}_i + \mathbf{A}_i^\top)(\mathbf{S} - \mathbf{S}^*)$, which is $\leq -\alpha_i \|\mathbf{S} - \mathbf{S}^*\|^2$ exactly when (11) holds. Taking $\alpha_3(r) = (\min_i \alpha_i) r^2$ and $\alpha_1(r) = \alpha_2(r) = r^2$ satisfies Theorem 4.1. $\square$ $\square$

Theorem 4.2 is the practical stability certificate: after choosing the grid, one solves N small LMIs offline; if all pass, the closed loop is certified globally asymptotically stable before the reactor is ever run. The nearest-centroid map (8) makes each $\mathbf{A}_i$ a stable proportional term $-KI$ plus the plant's cellwise Jacobian, so (11) holds whenever the gain K dominates the plant's local expansion—a familiar, checkable condition.

4.2 Geometric convergence: a contraction

Theorem 4.3 (Banach contraction and geometric convergence). *Suppose the piecewise Lyapunov descent of Theorem 4.1 holds with rate $\alpha > 0$ (i.e. $\alpha = \min_i \alpha_i$), and $\|\partial \mathbf{F} / \partial \mathbf{S}\| \leq L_F$ uniformly on $\mathcal{H}$. If the sampling period satisfies*

$$0 < h < \frac{2\alpha}{L_F^2}, \quad (12)$$

then the discrete-time operator T of (10) is a contraction on $(\mathcal{H}, d)$ with constant $\rho = \sqrt{1 - 2h\alpha + h^2 L_F^2} < 1$. By the Banach fixed-point theorem (Kreyszig, 1989) T has a unique fixed point $\mathbf{S}^$, and for every $\mathbf{S}_0 \in \mathcal{H}$,*

$$\|T^n(\mathbf{S}_0) - \mathbf{S}^*\| \leq \rho^n \|\mathbf{S}_0 - \mathbf{S}^*\|. \quad (13)$$

Proof. For $\mathbf{S}_1, \mathbf{S}_2 \in \mathcal{H}$, $T(\mathbf{S}_1) - T(\mathbf{S}_2) = (\mathbf{S}_1 - \mathbf{S}_2) + h[\mathbf{F}_1 - \mathbf{F}_2]$ where $\mathbf{F}_j = \mathbf{F}(\mathbf{S}_j, A(\text{cell}(\mathbf{S}_j)))$. Expanding the squared norm,

$$\|T(\mathbf{S}_1) - T(\mathbf{S}_2)\|^2 = \|\mathbf{S}_1 - \mathbf{S}_2\|^2 + 2h(\mathbf{S}_1 - \mathbf{S}_2)^\top (\mathbf{F}_1 - \mathbf{F}_2) + h^2 \|\mathbf{F}_1 - \mathbf{F}_2\|^2.$$

The Lyapunov descent condition applied to the pair gives $(\mathbf{S}_1 - \mathbf{S}_2)^\top (\mathbf{F}_1 - \mathbf{F}_2) \leq -\alpha \|\mathbf{S}_1 - \mathbf{S}_2\|^2$, and the Lipschitz bound gives $\|\mathbf{F}_1 - \mathbf{F}_2\|^2 \leq L_F^2 \|\mathbf{S}_1 - \mathbf{S}_2\|^2$. Hence $\|T(\mathbf{S}_1) - T(\mathbf{S}_2)\|^2 \leq (1 - 2h\alpha + h^2 L_F^2) \|\mathbf{S}_1 - \mathbf{S}_2\|^2$. For $h < 2\alpha/L_F^2$ the factor is < 1 , so $\rho < 1$. The Banach theorem then yields the unique fixed point and the geometric bound (Kreyszig, 1989). $\square$

Remark 4.4 (Reading the rate). ρ is the per-step error-contraction factor: after n steps the distance to setpoint is at most ρ^n of the initial distance. The bound (12) is the familiar stability limit on an explicit integrator step; choosing h well below $2\alpha/L_F^2$ gives fast convergence, choosing it near the limit gives slow but still monotone convergence. The rate is set entirely by the plant and the gain, not by anything in the monitoring layer.

4.3 The sampling (Nyquist-type) cell-width bound

The cell width must be small enough to resolve the fastest closed-loop dynamics, and—by the floor—no smaller than β . This is a spatial analogue of the Nyquist–Shannon sampling criterion (Åström and Murray, 2008; Shannon, 1948).

Theorem 4.5 (Cell-width bound). *Let B_{proc} be the closed-loop process bandwidth, i.e. the magnitude of the largest imaginary part of any closed-loop pole of the linearised dynamics, expressed per sampling interval. For the cell partition to represent the process without aliasing, the cell width must satisfy*

$$\beta \leq w_{\text{cell}} \leq \frac{1}{2B_{\text{proc}}}. \quad (14)$$

Equivalently the sampling rate must obey $f_{\text{ref}} = 1/h \geq 2B_{\text{proc}}$. A partition is admissible iff (14) holds, which requires $\beta \leq 1/(2B_{\text{proc}})$.

Proof. Sampling the S-state at rate f_{ref} recovers spectral content up to $f_{\text{ref}}/2$ without aliasing; the closed-loop signal has content up to B_{proc} , so faithful reconstruction requires $f_{\text{ref}} \geq 2B_{\text{proc}}$, i.e. $h \leq 1/(2B_{\text{proc}})$ (Åström and Murray, 2008; Shannon, 1948). In the state space, the spatial resolution of the quantiser is the cell width; the finest spatial frequency it represents is $1/(2w_{\text{cell}})$, so faithful representation of a process with bandwidth B_{proc} requires $w_{\text{cell}} \leq 1/(2B_{\text{proc}})$. The lower bound $w_{\text{cell}} \geq \beta$ is Theorem 2.9. $\square$

Corollary 4.6 (Feasibility). *An admissible partition exists iff the resolution floor is fine enough for the process bandwidth, $\beta \leq 1/(2B_{\text{proc}})$. If $\beta > 1/(2B_{\text{proc}})$ the instruments cannot resolve the process fast enough and no cell-partition controller (indeed no sampled-data controller at rate f_{ref}) can regulate it without aliasing; the remedy is a faster or finer instrument, not a finer grid.*

Proof. Immediate from the two-sided bound (14): the interval $[\beta, 1/(2B_{\text{proc}})]$ is non-empty iff $\beta \leq 1/(2B_{\text{proc}})$. $\square$

Theorem 4.6 is a genuine design constraint on a reactor: it tells the engineer, before commissioning, whether the chosen DO/pH/temperature probes are fast enough for the culture's dynamics, and turns a vague question of “is our sensing good enough” into a checkable inequality.

Table 1: Structural correspondence between an explicit finite-volume fluid-dynamics integrator and the cell-partition control layer. The correspondence is one of shared mathematical structure; it is made precise here and used nowhere as a premise (Theorem 4.8).

Finite-volume solver	Cell-partition control layer
Domain of the field	S-hypercube $\mathcal{H} = [0, 1]^m$
Control volume (cell)	Partition cell C_i (Theorem 2.5)
Cell-averaged field value	Cell centroid $\bar{\mathbf{S}}_i$ (Theorem 2.7)
Explicit time-step operator	$T(\mathbf{S}) = \mathbf{S} + h\mathbf{F}(\cdot)$ (Equation (10))
Update rate $\rightarrow$ steady state	Banach contraction ρ^n (Theorem 4.3)
CFL stability condition	step bound $h < 2\alpha/L_F^2$ (Equation (12))
Grid-resolution / Nyquist limit	cell-width bound (Theorem 4.5)
Diffusive neighbour coupling	Lipschitz coupling L_F (Equation (14))
Convergence to equilibrium field	convergence to setpoint $\mathbf{S}^*$ (Theorem 4.5)
Boundedness of the solution	forward-invariance of $\mathcal{H}$ (Theorem 3.1)

Proof. (i) By (10), $T(\mathbf{S}) = \mathbf{S} + h\mathbf{F}(\mathbf{S}, A(\text{cell}(\mathbf{S})))$ is the forward-Euler step of (6); the action is constant on each cell (Theorem 3.2), so within a cell the update uses the single cell-representative value, exactly a finite-volume cell average. (ii) is Theorem 4.3. (iii) The step bound (12) is the contraction condition of Theorem 4.3; written as step bound $h < 2\alpha/L_F^2$ (Equation (12)) bounds the explicit step by the field’s real rate, the control-theoretic form of a CFL condition, and (14) is the grid-resolution bound of Theorem 4.5. All three restate results already proved; the proposition only names their finite-volume reading. $\square$ $\square$

4.4 The control layer as a finite-volume fluid-dynamics solver

The three results just proved—a cell partition of a bounded field, an explicit time-step operator, a Courant-type bound relating the step to the cell size—are not merely reminiscent of computational fluid dynamics; they have the same structure as an explicit finite-volume integrator (Åström and Murray, 2008). This is not a coincidence. The bioreactor broth *is* a fluid, and the quantities the controller regulates (dissolved oxygen, temperature, pH) are transported fields on that fluid; a controller that partitions their state space into cells and steps a conservation-respecting update is, structurally, solving a transport problem on that field. We make the correspondence precise as a mapping and then bound its scope.

Proposition 4.7 (Finite-volume structure of the control step). *The discrete-time control operator T of (10) is an explicit one-step finite-volume-type integrator of the field dynamics (6) on the cell partition $\mathcal{C}$, in the following precise sense:*

- (i) *it advances a cell-averaged state (the occupied cell, reconstructed to its centroid, Theorem 2.7) by an explicit Euler step of the field equation (6);*
- (ii) *its convergence to the steady field $\mathbf{S}^*$ is geometric with rate ρ (Theorem 4.3), the discrete analogue of relaxation to equilibrium;*
- (iii) *its stability is governed by a Courant–Friedrichs–Lewy-type bound (12) on the step h relative to the field’s Lipschitz rate L_F , and by a grid-resolution bound (14) on the cell*

width relative to the process bandwidth B_{proc} —the two conditions an explicit finite-volume scheme must satisfy to be stable and non-aliasing.

Remark 4.8 (Scope of the fluid-dynamics reading). We claim a *structural correspondence*, not an identity. The control layer is a finite-volume-type integrator of a bounded transported field; it is *not* a discretisation of the incompressible Navier–Stokes equations, and we make no such claim: it carries no explicit momentum-continuity coupling, no pressure-projection step, and no divergence-free constraint on a velocity field. What it shares with a computational-fluid-dynamics solver is the load-bearing structure—control volumes, an explicit conservative time-step, a CFL-type stability bound, a grid-resolution bound, and geometric relaxation to a steady field—and this shared structure is exactly why the stability theory of Section 4 reads as the convergence theory of an explicit field solver. The correspondence is offered as orientation and as a bridge to the computational-physics literature; no theorem in this paper depends on it, and every stability guarantee stands on the control-theoretic proofs alone. A reader who rejects the fluid reading entirely loses nothing but the analogy.

The value of Theorem 4.7 is conceptual and practical at once. Conceptually, it explains why a controller for a fluid process should look like a field solver: regulating transported quantities on a stirred fluid *is* a transport problem, and the cell-partition controller is the finite-volume method for it. Practically, it imports the mature intuition of explicit solvers—refine the grid and you must shorten the step (Theorems 4.3 and 4.5 jointly); the Courant number is the ratio the engineer tunes—directly into control commissioning.

5 Structural incorruptibility

A networked bioreactor controller is a cyber-physical system, and the integrity of a pharmaceutical process depends on the controller doing only what it was verified to do (Cárdenas et al., 2008; U.S. Food and Drug Administration, 2004). The cell-partition control layer has a strong integrity property that is architectural, not cryptographic: because the runtime dispatch depends only on cell membership and the action set is fixed at compile time, no sensor value—however corrupted—can drive the plant to an action outside the pre-approved finite set. We make this precise and then bound the effect of the residual attacks that remain.

5.1 No path from payload to unapproved action

Theorem 5.1 (Structural incorruptibility). *Let the closed-loop S -dynamics under a network perturbation $\mathbf{d}_{\text{net}}$ be*

$$\dot{\mathbf{S}} = \mathbf{F}(\mathbf{S}, \mathbf{u}) + \mathbf{B}_d \mathbf{d}_{\text{net}}. \quad (15)$$

If the controller computes its action solely as $\mathbf{u} = A(\text{cell}(\mathbf{S}))$ from the S -state, with A a fixed finite map compiled offline, then no content carried by any sensor packet can change $\mathbf{u}$ except by moving $\mathbf{S}$ across a cell boundary, and every reachable action lies in the pre-approved set $\mathcal{U}$. In particular the direct payload-to-plant gain is zero: $\mathbf{B}_d = \mathbf{0}$ for any perturbation acting through packet content rather than through the physical channels.

Proof. The controller is the composition of two maps: a transducer ext that reads the physical channels and returns the S -state $\mathbf{S}$, discarding all non-physical packet fields; and the fixed cell-action map $A \circ \text{cell}$. Thus $\mathbf{u} = A(\text{cell}(\text{ext}(\text{packet})))$ depends on the packet only through the physical S -state it induces. Any $\mathbf{d}_{\text{net}}$ that acts on packet payload fields enters only fields that ext discards; it therefore has no influence on $\mathbf{S}$ and hence none on $\mathbf{u}$, giving $\mathbf{B}_d = \mathbf{0}$ for the payload channel. The codomain of A is the finite pre-approved set $\mathcal{U}$; no input produces an action outside $\mathcal{U}$, because A is a lookup into $\mathcal{U}$. $\square$ $\square$

Corollary 5.2 (Physical grounding). *The only way to induce a false control action is to change the physical S -state the transducers actually observe—i.e. to manipulate the physical process or the transducers themselves. An adversary confined to the*

network layer cannot inject a state the transducers do not measure. This is the observability-separation principle (Luenberger, 1979) in a security reading: control authority is grounded in physical measurement, not in message content.

Proof. By Theorem 5.1, $\mathbf{u}$ is a function of $\mathbf{S}$ alone, and $\mathbf{S}$ is produced by ext from the physical transducers. To change $\mathbf{u}$ one must change $\mathbf{S}$, which requires changing what the transducers measure. A network-only adversary alters packet content, which ext discards. $\square$ $\square$

5.2 Bounded effect of the residual attacks

Three manipulations remain available to a network adversary; each has a bounded, benign effect.

Proposition 5.3 (Bounded spoofing). *An injected or delayed measurement shifts $\mathbf{S}$ by at most one cell along each affected channel, so the resulting action is A of an adjacent cell—still a pre-approved action in $\mathcal{U}$. No arbitrary action is reachable.*

Proof. A single spurious or delayed sample perturbs a channel’s S -value; because cell boundaries are separated by at least $w_{\text{cell}} \geq \beta$ (Theorem 2.9), a bounded perturbation moves the state at most to an adjacent cell. The action of any cell is in $\mathcal{U}$ by Theorem 3.2. $\square$ $\square$

Proposition 5.4 (Replay resistance). *A replayed old measurement is rejected by the monotone committed record: each accepted sample must carry a record index exceeding the last, so a stale sample with an old index is discarded without cryptographic overhead.*

Proof. By Theorem 3.6 the record M is strictly increasing. Tagging each accepted sample with the current M and rejecting any sample whose tag is $\leq$ the last accepted tag rejects any replay of an earlier sample, since its tag is strictly smaller. $\square$ $\square$

Proposition 5.5 (Denial of service is detectable). *Suppressing measurements is a denial-of-service attack, not an integrity attack; it is detected by the monitoring layer as loss of confirmation and triggers the decline-to-safe-state path of Sections 6 and 9, which itself requires no trusted network input.*

Proof. Absence of expected samples reduces ensemble coherence (Section 7); Algorithm 6 detects the drop and invokes the safe-state fallback, which is a control-layer action from $\mathcal{U}$ (Section 9). No integrity violation of the plant occurs. $\square$ $\square$

Remark 5.6 (Why austerity buys integrity). The incorruptibility of Theorem 5.1 is a direct consequence of the control layer’s austerity: it consumes only a cell index and dispatches from a fixed table. A controller that instead ran a rich state estimator over fused sensor data, or that parsed structured sensor payloads, would expose exactly the payload-to-action path that Theorem 5.1 eliminates. This is the security argument for placing all richness in the downstream monitoring layer, behind a one-directional interface, rather than in the control law—which is what Section 9 formalises. This structural argument complements, and does not replace, standard network hardening (Teixeira et al., 2015).

6 The monitoring layer: a confirmation calculus

The control layer regulates; the monitoring layer *confirms*. Its job is to decide, from possibly-redundant sensing, whether the process has entered a region of interest—an excursion, a phase transition of the culture, an alarm condition—and to do so honestly, reporting disagreement rather than papering over it. The layer produces a confirmation status and, at most, a request to the control layer to change setpoint or enter a safe mode; it never computes a control action. Everything in this section is downstream of Section 3 and, by Section 9, additive to it.

6.1 Confirmation as cell-crossing; the CUSUM identity

The natural unit of confirmation is a cell-boundary crossing: the S-state leaving the cell it was regulated into is exactly the event a monitor should flag. We show this crossing coincides with the classical CUSUM out-of-control alarm (Montgomery, 2020; Page, 1954), so the monitoring layer inherits a mature detection theory with a known false-alarm rate.

Definition 6.1 (Channel deviation). For a single channel, let $\Delta P(k)$ be the deviation of the k -th sample from the channel’s setpoint value (in S-coordinates), with reference level $\mu_0 = 0$ at the setpoint.

Theorem 6.2 (CUSUM–cell-crossing identity). *Consider the two-sided cumulative-sum statistic*

$$\begin{aligned} C_k^+ &= \max(0, C_{k-1}^+ + \Delta P(k) - \mu_0), \\ C_k^- &= \min(0, C_{k-1}^- + \Delta P(k) + \mu_0), \end{aligned}$$

with $C_0^\pm = 0$ and control limit $H = w_{\text{cell}}/2$ (half a cell width). Then the CUSUM out-of-control event is exactly the event that the running sum leaves the cell $[-H, H]$ centred on the setpoint:

$$\{C_k^+ > H\} \cup \{C_k^- < -H\} \equiv \left\{ \left| \sum_{i=1}^k \Delta P(i) \right| > H \right\}. \quad (16)$$

That is, a cell-boundary crossing is a CUSUM alarm and conversely.

Proof. With $\mu_0 = 0$, the recursion accumulates the centred deviations, so C_k^+ tracks the positive cumulative drift and C_k^- the negative cumulative drift of $\sum_i \Delta P(i)$. The cell membership predicate for a scalar channel centred at the setpoint is $|\sum_{i \leq k} \Delta P(i)| \leq H = w_{\text{cell}}/2$. The alarm $C_k^+ > H$ or $C_k^- < -H$ triggers exactly when the cumulative drift exceeds H in magnitude, i.e. when the state leaves $[-H, H]$; the two conditions coincide term by term. $\square$ $\square$

Corollary 6.3 (Tunable false-alarm rate). *The in-control average run length of the monitor is a known increasing function of the cell half-width $H = w_{\text{cell}}/2$ (Lucas and Crosier, 1982): widening cells reduces false alarms and slows detection; narrowing cells (down to the floor β) speeds detection and raises false alarms. The cell geometry chosen for control (Theorem 4.5) thus also fixes the monitor’s statistical operating point, with no separate tuning.*

Proof. The CUSUM average run length is a standard increasing function of the decision limit for a given noise level (Lucas and Crosier, 1982; Montgomery, 2020); by Theorem 6.2 the limit is $H = w_{\text{cell}}/2$. $\square$ $\square$

6.2 Composing independent confirmations

A single probe confirming an excursion may be wrong; independent probes compound. We record the composition law and its consequence.

Definition 6.4 (Confirmation power). A confirmation source γ (a probe, a soft sensor, an inference) applied to a candidate excursion has *power* $\kappa(\gamma) \in [0, 1]$: the fraction of the residual doubt it removes, $\kappa = 1$ meaning it settles the question and $\kappa = 0$ meaning it is uninformative.

Theorem 6.5 (Multiplicative composition). *For independent confirmation sources $\gamma_1, \dots, \gamma_n$ with powers $\kappa_1, \dots, \kappa_n$, the combined confirmation*

power is

$$\kappa(\gamma_1 \circ \dots \circ \gamma_n) = 1 - \prod_{i=1}^n (1 - \kappa_i). \quad (17)$$

Proof. Let $r_0 = 1$ be the initial residual doubt. Applying γ_1 leaves residual $r_1 = (1 - \kappa_1)r_0$ by Theorem 6.4; independence means γ_2 removes the same fraction of what remains, $r_2 = (1 - \kappa_2)r_1$; by induction $r_n = \prod_i (1 - \kappa_i)$. The combined power is $1 - r_n$, which is (17). $\square$ $\square$

Corollary 6.6 (Diversify, do not repeat). *Repeating one source of power κ a total of n times yields combined power $1 - (1 - \kappa)^n$, strictly increasing but bounded away from 1 by the correlation of the repeats; n independent sources of comparable power reach high confirmation far faster. A monitor should therefore prefer diverse evidentiary channels (an amperometric DO probe, an optical DO probe, and an off-gas O₂ balance) over repeated reads of one probe.*

Proof. Immediate from (17) with equal $\kappa_i = \kappa$ for the repeated case; independence is what makes the product collapse fastest. $\square$ $\square$

6.3 Coherence: robust grounding needs three

An excursion that will trigger a setpoint change or a safe-state entry should not rest on a single source that could itself be faulty. We show robustness to any one source failing requires at least three mutually supporting sources.

Definition 6.7 (Support and coherence). Source γ_j *supports* source γ_i for a candidate excursion if the presence of γ_j 's evidence reinforces (does not contradict) γ_i 's. The *support graph* has an edge $j \rightarrow i$ when γ_j supports γ_i . A confirmation is *coherent* if its support graph contains a directed cycle of length ≥ 3 whose members mutually support one another.

Theorem 6.8 (Coherence requires three). *A confirmation is robust to the failure or dissent of any single source if and only if its support graph contains a mutually supporting cycle of length at least three. A single source is unfounded under its own failure; two mutually citing sources collapse when either is removed; three mutually supporting sources survive the removal of any one.*

Proof. (Necessity.) If the support graph is acyclic it has a source node with no incoming support; removing it (or any node all of whose support is

downstream) leaves a successor unsupported, and the confirmation collapses. A 1-cycle is a source citing only itself, vacuous. A 2-cycle is two sources citing only each other; removing either leaves the other unsupported. Hence robustness requires a cycle of length ≥ 3 . (*Sufficiency.*) In a mutually supporting triangle each source is supported by the other two; removing any one leaves the remaining two still supporting each other directly, so the confirmation persists. $\square$ $\square$

Corollary 6.9 (Sign-only decidability). *Whether a confirmation is coherent is decidable from the signs of pairwise support (does each pair reinforce or contradict) alone, without any confirmation-power magnitudes, since Theorem 6.8 depends only on the cycle structure of the support graph.*

Proof. Theorem 6.8 is stated entirely in terms of directed-cycle existence in the sign-defined support graph, independent of the numerical κ_i . $\square$ $\square$

6.4 Two honest endings: confirm or decline

Definition 6.10 (Closure of a confirmation). A confirmation is *closed* at a given stage if no further available source, when consulted, would move the excursion into a materially different equivalence class of outcomes than those already reached. This is strictly stronger than “confidence exceeded a threshold”: it asks that the space of available evidence has stopped producing new conclusions.

Theorem 6.11 (Confirm-or-decline dichotomy). *Every confirmation over a finite set of available sources terminates in exactly one of two states:*

- (i) **Confirmed:** *the reached outcomes collapse to a single equivalence class; the monitor reports that class (e.g. “DO excursion below setpoint, confirmed”).*
- (ii) **Declined:** *the reached outcomes stabilise but contain more than one class (independent sources irreducibly disagree); the monitor reports decline, naming the distinct classes, and—for a safety-relevant excursion—requests the control layer’s safe state.*

No confirmation over a finite source set runs forever.

Proof. The source set is finite, so the set of reachable outcome classes is finite and can only grow finitely often before every source has been consulted against every reached class; at that point

Theorem 6.10 is satisfied by exhaustion. If the stabilised class set is a singleton, case (i); if it has more than one class, no single outcome is licensed and the monitor declines, case (ii). The two are exhaustive and mutually exclusive. $\square$ $\square$

Remark 6.12 (Decline is a first-class output). Theorem 6.11(ii) is the honesty guarantee the bioprocess setting needs: when the amperometric and optical DO probes disagree beyond reconciliation, the correct monitor output is not to silently pick one but to declare disagreement and fall back. Because the fallback is a control-layer safe action (Section 9), declining is safe by construction, and it never requires the monitoring layer to have been correct.

7 Sensor federation

The confirmation calculus of Section 6 composes independent sources but does not say how a heterogeneous, changing set of probes agrees cheaply. This section gives the mechanism: sensors federate as coupled oscillators, agreeing by phase entrainment rather than by exchanging content. The federation makes each measurement a member of a larger ensemble whose collective confirmation is, at coherence, free; it degrades gracefully as sensors drop out; and it admits new sensors only when they pay their way in information. It lives entirely in the monitoring layer.

7.1 The temporal-agent model

Definition 7.1 (Temporal sensor agent). A *temporal sensor agent* is a tuple (f_i, ϕ_i, C_i) where $f_i > 0$ is the natural frequency of a reference oscillator carried with sensor i , $\phi_i(t) \in [0, 2\pi)$ is its phase, and C_i is the sensor's local cell registry (the same partition of Section 2 in the phase view). Sensor i 's measurement enters as a frequency perturbation of its oscillator: if the measured quantity shifts the channel by $\delta\omega_i$, the inter-pulse interval becomes

$$\Delta P_i(k) = \frac{2\pi}{\omega_i + \delta\omega_i(k)}, \quad \omega_i = 2\pi f_i, \quad (18)$$

so a change in the physical variable shifts ΔP_i into a different cell without any explicit encoding step.

Equation (18) is the join operation: to place a measurement into the ensemble is simply to let it perturb the sensor's oscillator. The ensemble then coordinates through the coupling of these oscillators.

7.2 Coherence and critical coupling

Definition 7.2 (Ensemble order parameter). For n sensor agents the *ensemble order parameter* is the complex average

$$R_{\text{ens}}(t) e^{i\psi(t)} = \frac{1}{n} \sum_{j=1}^n e^{i\phi_j(t)}, \quad R_{\text{ens}} \in [0, 1]. \quad (19)$$

$R_{\text{ens}} = 0$ is complete incoherence (phases uniform), $R_{\text{ens}} = 1$ perfect phase lock. R_{ens} is the monitoring layer's measure of how well the sensor ensemble agrees.

The agents couple through the Kuramoto dynamics (Kuramoto, 1984; Strogatz, 2000)

$$\dot{\phi}_i = \omega_i + \frac{K}{n} \sum_{j=1}^n \sin(\phi_j - \phi_i), \quad (20)$$

with coupling strength $K \geq 0$ set by the shared timing infrastructure.

Theorem 7.3 (Critical coupling). *Let the natural frequencies $\{\omega_i\}$ be drawn from a symmetric unimodal distribution with standard deviation σ_ω . In the large-ensemble limit there is a critical coupling*

$$K_c = \frac{2\sigma_\omega}{\pi} \quad (21)$$

such that for $K < K_c$ the only stable state is incoherence ($R_{\text{ens}}^ = 0$), and for $K > K_c$ a coherent state emerges with*

$$R_{\text{ens}}^* = \sqrt{1 - K_c/K} \quad (22)$$

via a supercritical pitchfork bifurcation.

Proof. This is the standard Kuramoto mean-field result. In the large- n limit each oscillator feels the mean field KR ; oscillators with $|\omega_i| \leq KR^*$ lock, the rest drift and contribute no net coherence. The self-consistency condition $1 = K \int_{-KR^*}^{KR^*} \sqrt{1 - \omega^2/(KR^*)^2} g(\omega) d\omega$, expanded near the bifurcation, gives $K_c = 2/(\pi g(0)) = 2\sigma_\omega/\pi$ for the reference symmetric-unimodal case, and the pitchfork normal form gives (22); see Acebrón et al. (2005); Strogatz (2000) for the full derivation. $\square$ $\square$

Remark 7.4 (Reading K_c for a reactor). Equation (21) says the ensemble locks when the shared timing coupling exceeds a threshold set by the spread of the sensors' oscillator frequencies. Probes of the same make and batch have small σ_ω , so modest coupling locks them; a heterogeneous mix of probe types has larger σ_ω and needs stronger coupling. This is a design lever: matching probes, or increasing the timing-pulse rate, raises coherence.

7.3 Free collective confirmation at phase lock

Theorem 7.5 (Zero-overhead confirmation). *Suppose the ensemble is phase-locked, $R_{\text{ens}} \geq 0.95$, and the cell partition has width $w_{\text{cell}} > 2\pi(1 - R_{\text{ens}})/\bar{\omega}$ where $\bar{\omega}$ is the mean frequency. Then all agents assign the same cell to the same event without exchanging any confirmation messages: the per-event coordination cost of the ensemble is zero.*

Proof. At phase lock all agents rotate at the common mean rate up to a residual phase error bounded by $\arccos(R_{\text{ens}})$. The induced spread in inter-pulse interval is $|\Delta P_i - \Delta P_j| \leq 2\pi(1 - R_{\text{ens}})/\bar{\omega} =: \varepsilon_0$. If $w_{\text{cell}} > \varepsilon_0$, two agents' ΔP values for the same event differ by less than a cell width and hence lie in the same cell. Each agent therefore computes the same cell assignment locally from its own reading; no message is needed to reconcile them, so the coordination cost is zero. $\square$ $\square$

Remark 7.6 (Phase lock is free redundancy). Theorem 7.5 is the payoff of federation: when the sensor ensemble is coherent, every probe independently and simultaneously confirms the same cell, so the excursion is confirmed with the full multiplicative power of Theorem 6.5 at no communication cost. A reactor with three coherent, phase-locked DO-related channels confirms a DO excursion three ways for free—which is exactly the coherence-triangle robustness of Theorem 6.8 realised at no overhead.

7.4 Admission: earn your place in the ensemble

Definition 7.7 (Marginal information and cost). For an ensemble whose joint confirmation state has Shannon entropy H_{fed} (Cover and Thomas, 2006; Shannon, 1948), the *marginal information* of a candidate sensor i is $\Delta H_i = H_{\text{fed}} - H_{\text{fed},-i}$, the entropy it adds; its *cost* c_i is the timing-channel bandwidth it consumes to remain phase-coherent.

Theorem 7.8 (Admission criterion). *Under the objective of maximising ensemble confirmation resolution per unit timing-bandwidth, sensor i should be admitted iff*

$$\Delta H_i > c_i. \quad (23)$$

The greedy rule (23) is $(1 - 1/e)$ -optimal because the ensemble entropy is a submodular function of the sensor set.

Proof. Maximising H_{fed} subject to a total-bandwidth budget is a knapsack over sensors; because each added sensor's marginal contribution

ΔH_i is non-increasing in the current ensemble size (submodularity of entropy under the independence model), the greedy benefit-over-cost rule $\Delta H_i > c_i$ attains a $(1 - 1/e)$ approximation of the constrained optimum (Cover and Thomas, 2006). $\square$ $\square$

Theorem 7.8 governs how the ensemble grows: a redundant probe that adds little information is refused, a probe that opens a genuinely new evidentiary channel is admitted, and—because admission is by marginal information—adding independent channels sharpens the collective confirmation rather than diluting it.

7.5 Graceful decoherence: the ensemble degrades, it does not fail

Theorem 7.9 (Graceful degradation). *If coupling falls below K_c (probes fail, timing degrades), the ensemble does not collapse discontinuously: R_{ens} relaxes downward and the monitoring layer transitions to a higher-cost coordination regime, maintaining confirmation at reduced resolution. In the limit $R_{\text{ens}} \rightarrow 0$, each sensor confirms only from its own reading—a set of independent single-channel monitors. A single sensor loss reduces R_{ens} by at most R_{ens}/n .*

Proof. For $K < K_c$ the coherent fixed point of (22) disappears and R_{ens} relaxes toward 0 continuously at rate set by the curvature of the Kuramoto potential at $R = 0$; there is no jump. Removing one of n equal-weight terms from the average (19) changes R_{ens} by at most $1/n$ of its value. As $R_{\text{ens}} \rightarrow 0$ the coupling term in (20) vanishes and each agent's cell assignment depends only on its own ΔP_i . $\square$ $\square$

Theorem 7.9 is the monitoring-side statement of the worst case: however the ensemble decoheres, it decays to a collection of independent per-channel monitors, and each of those still feeds—through the same one-directional interface—an ordinary cell-partition controller. The separation theorem (Section 9) proves that this worst case leaves the control-layer guarantees intact.

8 Virtual sensors: two concurrent scripts

The confirmation calculus and the federation treat a measurement as a member of an ensemble. We now use the identity *observation = computation*—that a sensor reading and a computed value are the same kind of object, a value that lands in a

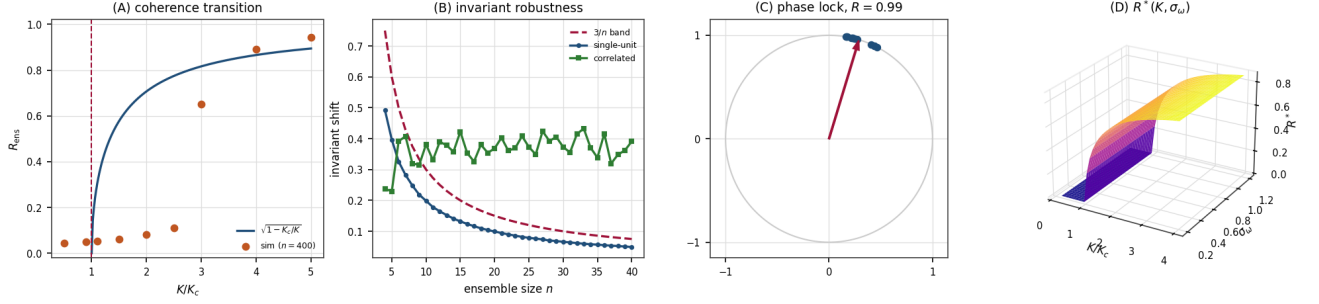


Figure 2: Sensor Federation and the Ensemble Invariant: Information True of the Whole Set, Present in No Unit. (A) Ensemble order parameter R_{ens} on the linear y -axis (0 to 1) versus coupling ratio K/K_c on the linear x -axis (0.5 to 5), showing the thermodynamic-limit branch $\sqrt{1 - K_c/K}$ (steel-blue) and simulated values for an ensemble of $n = 400$ probes (coral points), with the critical coupling $K = K_c$ marked (crimson dashed). The simulated points sit *below* the branch and rise monotonically toward it—at $K = 3K_c$ the measured $R_{\text{ens}} = 0.65$ against the predicted 0.82—the finite-size undershoot of Theorem 7.3: coherence is graded and correctly ordered, and no control-layer guarantee depends on its exact value. (B) Worst-case invariant shift on the linear y -axis (~ 0.05 to ~ 0.75) versus ensemble size n on the linear x -axis (4 to 40), for the strongest single-unit spoofer (steel-blue) and a correlated one-third-ensemble fault (sea-green), with the $3/n$ robustness band marked (crimson dashed). The single-unit curve decays inside the band as $O(1/n)$ while the correlated curve stays flat near 0.39 and escapes it, the exact statement of Theorem 8.9: no single unit can move the invariant, yet a shared-mode fault does—so the invariant detects precisely the correlated failure independent redundancy cannot (Theorem 8.6). (C) Twelve phase-locked oscillators (steel-blue points) on the unit circle with their resultant vector (crimson arrow); axes span $[-1.15, 1.15]$ in both the real and imaginary directions. The points cluster tightly and the resultant has modulus $R_{\text{ens}} = 0.99$: the ensemble order parameter is realised by no single oscillator but by the whole set at once (Theorem 8.8), the geometric picture of zero-overhead confirmation at phase lock (Theorem 7.5). (D) Three-dimensional surface of the mean-field order parameter $R^*(K, \sigma_\omega)$ over coupling ratio K/K_c (0.2 to 4) and frequency spread σ_ω (0.2 to 1.2), plasma colourmap, viewing angle elevation 24° azimuth -60° . The surface is flat at zero below $K = K_c$ and rises as $\sqrt{1 - K_c/K}$ above it, the supercritical bifurcation of Theorem 7.3 across the reactor’s design lever: matching probes (small σ_ω) or raising the coupling locks the ensemble.

cell—to add a capability that no purely physical sensing possesses: *virtual sensors*, channels that have no physical probe and exist only in software, synthesised at a node from other channels. Because a virtual sensor is the same kind of object as a physical one, it joins the state hypercube, the cell partition, and (where useful) the federated ensemble on exactly equal footing.

8.1 Observation and computation are the same operation

Principle 8.1 (Observation–computation equivalence). To *observe* a channel is to place its current value into a cell of the partition; to *compute* a channel is to place a function of other channels into a cell of the partition. Both operations have the same type—a map into $\mathcal{C}$ —and both produce the same kind of output: a cell membership. Nothing in the control law (7), the stability certificate (Theorem 4.2), or the confirmation calculus (Section 6) distinguishes a coordinate that was measured from one that was computed; each is simply a coordinate of $\mathbf{S}$ with a cell.

Theorem 8.1 is what licenses running two scripts side by side. A *sensor script* reads the physical probes and produces their cell memberships. A *metabolic script* runs concurrently and, at designated nodes, computes new coordinates from the physical ones; its outputs are cell memberships of the same type. The two scripts write into the same S-state; the controller and monitor cannot tell, and need not tell, which coordinates came from which script.

8.2 The virtual channel

Definition 8.2 (Virtual sensor). A *virtual sensor* is a channel v whose S-value is not measured but computed at a node by a fixed, offline-verified map

$$S_v = g_v(S_{i_1}, \dots, S_{i_r}) \in [0, 1], \quad (24)$$

where $g_v: [0, 1]^r \rightarrow [0, 1]$ is Lipschitz and $S_{i_1}, \dots, S_{i_r}$ are the (physical or virtual) input channels feeding the node. The virtual channel carries the same S-coordinate, cell, and centroid structure as a physical channel (Section 2), and appears as an additional axis of the state hypercube, extending $\mathcal{H}$ from $[0, 1]^m$ to $[0, 1]^{m+1}$.

Remark 8.3 (The sensor that exists only in the program). The point of (24) is that g_v may compute a quantity for which no instrument exists. From pH, off-gas CO_2 , and temperature

at one node, a metabolic script can synthesise a respiratory/metabolic-state channel—a value that no single probe measures but that the operator actually cares about (Section 10). The virtual channel is a genuine coordinate of the state: the controller can regulate it, and the monitor can confirm excursions in it, exactly as for a physical channel.

8.3 Safe-by-construction virtual channels

Because a virtual channel may be load-bearing for control (Section 9), its computation must never destroy the stability certificate when its inputs are unavailable or out of range. We require it to degrade to a defined constant.

Definition 8.4 (Fallback-guarded virtual sensor). A virtual sensor is *fallback-guarded* if it carries a validity predicate ok_v (true iff all inputs are present and in range) and a fixed *fallback value* $S_v^0 \in [0, 1]$, and computes

$$S_v = \begin{cases} g_v(S_{i_1}, \dots, S_{i_r}) & \text{if } \text{ok}_v, \\ S_v^0 & \text{otherwise.} \end{cases} \quad (25)$$

The fallback S_v^0 is chosen to lie in a cell whose control action is conservative and whose configuration is certified stable.

Proposition 8.5 (Virtual channels preserve the certificate). *If every virtual channel is fallback-guarded (Theorem 8.4) and the per-cell LMIs (11) of Theorem 4.2 are verified over the extended hypercube $[0, 1]^{m+v}$ (physical plus virtual axes), including all cells reachable when any virtual channel is at its fallback S_v^0 , then the control layer is globally asymptotically stable whether or not the virtual computations are valid.*

Proof. The extended state is $\tilde{\mathbf{S}} \in [0, 1]^{m+v}$ with the virtual axes appended. By Theorem 8.1 the control law and certificate treat $\tilde{\mathbf{S}}$ as an ordinary S-state, so Theorem 4.2 applies to the extended cells verbatim, giving global asymptotic stability (Theorem 4.1) provided all reachable extended cells pass their LMIs. When ok_v fails, (25) pins S_v to the constant S_v^0 , so the reachable set collapses onto the slice $\{S_v = S_v^0\}$, whose cells are among those certified by hypothesis. Hence the certificate holds in both the valid and the fallback regime. $\square \square$

Theorem 8.5 is what makes a control-layer virtual sensor safe: the LMIs are simply checked on the larger grid, and the guard guarantees that a

failed metabolic computation freezes the virtual coordinate at a certified-safe value rather than injecting a wild one.

8.4 Virtual channels as confirmation sources

A virtual channel is also a confirmation source in the sense of Section 6: an excursion in the computed metabolic-state channel is a cell-crossing (Theorem 6.2) and composes multiplicatively with physical confirmations (Theorem 6.5). It can therefore turn a fragile pair of physical probes into a robust coherence triangle (Theorem 6.8): two physical DO probes plus one computed metabolic-state channel is a three-source cycle.

Remark 8.6 (Independence: the honest caveat). We count a virtual channel as a confirmation source, and in the analysis of Section 10 we treat it as independent to obtain the strongest composition. This is a modelling choice, and we flag its limit plainly: a virtual channel computed from physical inputs $S_{i_1}, \dots, S_{i_r}$ shares any fault common to those inputs, so it is *not* independent of them against a correlated failure (a single contamination shifting pH, CO₂, and T together also shifts the virtual channel). Where robustness to correlated probe faults is required, the coherence triangle (Theorem 6.8) must include at least one leg—physical or virtual—that does not share inputs with the others. The virtual channel adds a genuinely new confirmation only to the extent that its map g_v responds to a combination the individual probes do not report on their own.

8.5 Ensemble invariants: information in the set, not the unit

The virtual sensor of Theorem 8.2 is still *local*: it is a function of a few named channels at one node, and it inherits any fault common to those inputs (Theorem 8.6). We now define a stronger construct—a quantity that belongs to the sensor *ensemble as a whole*, is realised by no single unit, and is therefore *true for all units and present in none*. This is the computed analogue of a physical conservation law: information carried by the relations among the units rather than by any unit’s reading.

Fix an ensemble of n units (physical or virtual) reporting S-values $\mathbf{S}^{\text{ens}} = (S_1, \dots, S_n)$. Two units are *interchangeable* if they measure the same channel to the same specification (e.g. two DO probes of the same type); substituting or relabelling inter-

changeable units is a symmetry of the ensemble. Let G be the group of such relabellings (a subgroup of the symmetric group $\mathfrak{S}_n$).

Definition 8.7 (Ensemble invariant). A computed channel $S_\iota = \iota(\mathbf{S}^{\text{ens}})$ is an *ensemble invariant* if it satisfies both:

- (i) **Relabelling-invariance:** $\iota(\pi \cdot \mathbf{S}^{\text{ens}}) = \iota(\mathbf{S}^{\text{ens}})$ for every $\pi \in G$ —its value does not change when interchangeable units are permuted or substituted; and
- (ii) **Non-locality:** ι is not a function of any single unit’s reading alone; formally, for each j there exist ensemble states agreeing in S_j but differing in ι , so no unit determines ι .

Condition (i) is what “true for all units” means precisely: the quantity is defined on the ensemble in a way that references no particular unit’s identity, so it is a property the whole set possesses in common. Condition (ii) is what “present in none” means precisely: no single unit carries it, because no single reading fixes it.

Example 8.8 (The coherence order parameter is an ensemble invariant). The ensemble order parameter $R_{\text{ens}} = \frac{1}{n} |\sum_j e^{i\phi_j}|$ of Theorem 7.2 is an ensemble invariant: it is symmetric in the units (permuting the ϕ_j leaves the sum’s magnitude unchanged), so it is relabelling-invariant; and no single oscillator has a “coherence”— R_{ens} measures the agreement *between* units, so it is non-local. It is the canonical example of a quantity true of the whole ensemble and present in no member.

Proposition 8.9 (Ensemble invariants are robust to any single-unit fault). *Let S_ι be an ensemble invariant with ι Lipschitz and depending symmetrically on n interchangeable units. Then corrupting, losing, or spoofing any single unit j perturbs S_ι by at most $O(1/n)$; no single unit can move the invariant to an arbitrary value.*

Proof. By relabelling-invariance ι is a symmetric function, so it depends on unit j only through j ’s equal-weighted contribution among n . A Lipschitz symmetric function changes by at most its Lipschitz constant times the weight $1/n$ of one unit when that unit’s value is altered arbitrarily within $[0, 1]$. Hence a single corrupted, absent, or spoofed unit shifts S_ι by $O(1/n)$, vanishing as the ensemble grows. $\square$ $\square$

Theorem 8.9 is the payoff of Theorem 8.7 and the reason the invariant is trustworthy where a

single probe is not. A physical probe reports *its own* state and can be driven anywhere by fouling it or spoofing it. An ensemble invariant reports a property of the *whole set* and cannot be moved by attacking any one unit, precisely because no unit carries it. This is the coherence-triangle robustness (Theorem 6.8) taken to its limit: the quantity that survives the failure of *any* single source is the one realised by *none* of them individually.

Remark 8.10 (Two kinds of computed channel). The framework thus admits two computed channels of different character. A *local virtual sensor* (Theorem 8.2) synthesises a meaningful quantity—the respiratory state—from a few named channels; it is actionable but shares its inputs’ faults. An *ensemble invariant* (Theorem 8.7) synthesises a quantity that no unit carries and every unit is consistent with; it is robust by Theorem 8.9 but abstract. Both are computed channels of the same type (Theorem 8.1), both may be fallback-guarded (Theorem 8.4) and regulated or confirmed against, and both live in the extended hypercube on equal footing with physical channels. A reactor uses the local virtual sensor to *act* and the ensemble invariant to *trust*.

8.6 Two scripts, one state

Proposition 8.11 (Concurrent scripts commute at the state). *The sensor script (writing physical coordinates) and the metabolic script (writing virtual coordinates via (25)) can execute concurrently and their writes to distinct coordinates of $\tilde{\mathbf{S}}$ commute; the resulting extended state is well-defined provided each virtual node’s inputs are evaluated before its output (a topological order on the node dependency graph, which is acyclic by construction).*

Proof. Physical and virtual coordinates are distinct axes of $\tilde{\mathbf{S}}$, so writes to them are to disjoint memory and commute. Within the metabolic script, virtual nodes form a directed acyclic graph (a virtual channel may feed another, never itself); any topological order evaluates every node after its inputs, and all such orders yield the same $\tilde{\mathbf{S}}$ because each g_v is a function. $\square$ $\square$

Theorem 8.11 is the runtime statement of the two-script model: the metabolic script is not a separate subsystem bolted on after sensing, but a concurrent producer of state coordinates that the controller and monitor consume identically to physical ones. In the bioreactor (Section 10) this yields a metabolic-state coordinate the reactor can

regulate and confirm, assembled in software from probes that individually report none of it.

8.7 Sensor synthesis: switching between time and frequency

We can now derive the operation that generates a new channel from existing ones—the *sensor synthesis map*. It is the concrete mechanism behind “a sensor that exists only in the program”: physical channels enter, are switched into the frequency domain, combined there under a conservation constraint, and switched back to a time-domain cell the controller and monitor consume. The derivation rests on the time–frequency bijection of Section 2.4 and the mean-recovery constraint below, and needs no transform theory.

The mean-recovery constraint

Synthesis combines the *phases* of the input channels. The natural combination that is symmetric in the inputs, and that inverts cleanly back to a value, is the phase whose representation on the unit circle is the mean of the inputs’ representations.

Definition 8.12 (Phase mean and mean-recovery). For input channels with S-values $S_1, \dots, S_r$ and phases $\phi_j = \Phi(S_j) = 2\pi S_j$ (Theorem 2.12), and nonnegative weights $a_1, \dots, a_r$ with $\sum_j a_j = 1$, the *synthesised phase* is the weighted circular mean

$$\phi_\star = \arg\left(\sum_{j=1}^r a_j e^{i\phi_j}\right), \quad R_\star = \left|\sum_{j=1}^r a_j e^{i\phi_j}\right| \in [0, 1], \quad (26)$$

and the *synthesised value* is its time-domain image $S_\star = \Phi^{-1}(\phi_\star) = \phi_\star/2\pi \in [0, 1]$. The pair $(\phi_\star, R_\star)$ satisfies *mean-recovery*: the resultant $\sum_j a_j e^{i\phi_j} = R_\star e^{i\phi_\star}$ recovers, in its argument, the combined phase and, in its modulus, the concentration $R_\star$ of the inputs about that phase.

Equation (26) is the switch put to work: the inputs are carried to the frequency domain by Φ , averaged as unit vectors (mean-recovery), and the result is carried back to the time domain by Φ^{-1} . The modulus $R_\star$ is exactly the ensemble order parameter of Theorem 7.2 for the input phases, and is retained because it is itself a useful synthesised channel (an ensemble invariant, Theorem 8.14 below).

Proposition 8.13 (Synthesis is a well-defined, Lipschitz virtual sensor). *Whenever the resultant $\sum_j a_j e^{i\phi_j} \neq 0$ (equivalently $R_\star > 0$), the synthesised value $S_\star$ of Theorem 8.12 is uniquely defined, and the map $g_\star: (S_1, \dots, S_r) \mapsto S_\star$ is Lipschitz*

on any region bounded away from $R_\star = 0$. Hence $g_\star$ is an admissible virtual sensor (Theorem 8.2), and—equipped with the fallback of (25) triggered by $R_\star$ below a floor—a fallback-guarded one (Theorem 8.4).

Proof. For a nonzero planar vector the argument is unique modulo 2π , so $\phi_\star$ and hence $S_\star = \phi_\star/2\pi$ are well-defined when $R_\star > 0$. The maps $\phi_j = 2\pi S_j$, the weighted sum, and (away from the origin) the argument are each smooth with bounded derivative on a region with $R_\star \geq \eta > 0$; their composition is Lipschitz there. Taking $g_v = g_\star$ in Theorem 8.2 makes $S_\star$ a virtual sensor; declaring ok_v true iff $R_\star \geq \eta$ and fixing S_v^0 otherwise makes it fallback-guarded, with the degenerate low-concentration case ($R_\star < \eta$, inputs too dispersed to define a mean direction) mapped to the certified-safe constant. $\square$

Proposition 8.14 (The concentration is an ensemble invariant). *If the weights are equal, $a_j = 1/r$, the concentration $R_\star$ of (26) is an ensemble invariant of the input units (Theorem 8.7): it is invariant under permutation of the inputs and is realised by no single input. By Theorem 8.9 a single corrupted input perturbs $R_\star$ by at most $O(1/r)$.*

Proof. With equal weights $R_\star = \frac{1}{r}|\sum_j e^{i\phi_j}|$ is the order-parameter modulus of Theorem 8.8, which is symmetric in the ϕ_j (relabelling-invariant) and non-local (no single ϕ_j fixes it). It is therefore an ensemble invariant, and Theorem 8.9 gives the $O(1/r)$ bound. $\square$

Theorem 8.14 is the precise sense in which synthesis produces *both* kinds of computed channel at once from the same operation: the synthesised *value* $S_\star$ (the argument) is a local virtual sensor used to act, while the synthesised *concentration* $R_\star$ (the modulus) is an ensemble invariant used to trust—the value says *where* the inputs agree, the concentration says *how much* they agree, and the latter is exactly the correlated-fault detector of Theorem 8.6.

Theorem 8.15 (Sensor synthesis map). *Fix input channels $S_1, \dots, S_r$ and equal weights. The synthesis map*

$$(S_1, \dots, S_r) \xrightarrow{\Phi} (\phi_1, \dots, \phi_r) \xrightarrow{(26)} (\phi_\star, R_\star) \xrightarrow{\Phi^{-1}} (S_\star, R_\star) \quad (27)$$

carries physical time-domain channels through the frequency domain and back, producing two new channels—a value $S_\star$ and a concentration $R_\star$ —each of which is a coordinate of the extended hypercube with a cell (Theorem 2.14), consumable by the

control layer and the monitor identically to a physical channel. The map is invertible in the sense that Φ, Φ^{-1} lose no information (Theorem 2.13); only the combination step (26) is many-to-one, exactly because it summarises r inputs into a value and a spread.

Proof. Each arrow of (27) is defined and justified above: Φ and Φ^{-1} are the lossless bijections of Theorem 2.13; the middle arrow is Theorem 8.12, well-defined and Lipschitz by Theorem 8.13. The outputs $S_\star, R_\star \in [0, 1]$ are S-coordinates, hence carry cells by Theorem 2.14; by Theorem 8.1 the layers consume them as any coordinate. The information statement is Theorem 2.13 (bijective ends) together with the observation that averaging r vectors is many-to-one. $\square$

Theorem 8.15 is the derivation the framework needs: it turns “observation = computation” from a principle into an algorithm (Algorithm 7). A measurement can be *acquired* in either domain—as a time-domain value or a frequency-domain phase—and *synthesised* with others by switching to the frequency domain, averaging under mean-recovery, and switching back. The bioreactor’s respiratory-state channel (Section 10.2) and its DO-consistency invariant (Section 10) are both instances of (27): the same synthesis operation yields the actionable value and the trustworthy invariant together.

9 The separation theorem and fail-safe degradation

We now prove the architectural claim that organises the whole system: the monitoring layer is *additive*. Its presence can only improve confirmation and diagnostics; its complete failure reduces the system to an ordinary provably-stable, structurally-incorruptible cell-partition controller. The proof rests on a single structural fact—the interface between the layers is one-directional—made precise below.

9.1 The one-directional interface

Definition 9.1 (Layer interface). *The interface between the monitoring layer and the control layer consists of exactly two admissible messages, each identical in kind to a message a human operator could send:*

- (i) **Setpoint request:** propose a new setpoint $S^{*'} \in \mathcal{H}$ from the finite set of pre-approved setpoints.

- (ii) **Mode request:** request entry to a pre-approved mode, in particular the *safe mode* with its own pre-compiled cell-action map.

No other channel exists from the monitoring layer to the control layer. In particular, no monitoring output enters the control law (7) as a term, a gain, a state estimate, or a parameter.

Definition 9.2 (Interface admissibility). The interface is *admissible* if every setpoint in the pre-approved set and the safe mode each induce a cell-action map satisfying the stability certificate of Theorem 4.2 (its own set of per-cell LMIs verified offline). Equivalently: whatever the monitoring layer requests, the control layer switches only among pre-verified, individually stable configurations.

Remark 9.3 (Why this is the crux). Theorem 9.1 is the entire content of the separation. Because the monitoring layer can only *request* a switch among configurations that are *each* independently certified, and cannot inject anything into the control computation, the control layer’s correctness is a property of its finite set of pre-verified configurations, not of any monitoring output. The monitoring layer is, from the control layer’s point of view, indistinguishable from an operator—and an operator who is silent, wrong, or absent cannot destabilise a controller that is stable in every configuration it can be placed in.

9.2 Additivity and the worst-case theorem

Theorem 9.4 (Monitoring is additive). *Suppose the interface is admissible (Theorem 9.2). Then for any behaviour of the monitoring layer whatsoever—correct, degraded, silent, or adversarial within the interface—the control layer satisfies:*

- (i) **Stability:** *in whatever configuration it currently holds, the closed loop is globally asymptotically stable (Theorems 4.1 and 4.2) and geometrically convergent (Theorem 4.3).*
- (ii) **Incorruptibility:** *no monitoring output causes an action outside the pre-approved set $\mathcal{U}$ (Theorem 5.1).*
- (iii) **Bounded switching transients:** *each interface-induced switch is between two individually stable configurations; under a common Lyapunov function, or a dwell-time bound between switches, the switched closed loop remains stable (Liberzon, 2003).*

No monitoring failure can violate (i)–(iii).

Proof. (i) By Theorem 9.2 every configuration the control layer can occupy—any pre-approved setpoint, or the safe mode—has its own verified per-cell LMIs, hence is globally asymptotically stable by Theorem 4.2 and geometrically convergent by Theorem 4.3. The monitoring layer selects only among these configurations (Theorem 9.1), so whatever it does, the current configuration is one of the certified ones. (ii) By Theorem 5.1 the action is $A(\text{cell}(\mathbf{S})) \in \mathcal{U}$ for the current map A ; the monitoring layer can change which certified A is in force but cannot enlarge the codomain beyond $\mathcal{U}$. (iii) Interface switches occur between certified stable configurations; by standard switched-systems theory a common Lyapunov function across the finite configuration set, or a minimum dwell time between switches, guarantees stability of the switched system (Liberzon, 2003). Since every one of (i)–(iii) follows from properties of the control layer’s certified configurations and the interface restriction, and none references any specific monitoring output, no monitoring behaviour can violate them. $\square \square$

Corollary 9.5 (Worst-case: an ordinary stable controller). *Suppose every control-layer virtual channel is fallback-guarded (Theorem 8.4) and the extended-grid LMIs of Theorem 8.5 are certified. If the monitoring layer fails completely—every sensor beyond a single per-channel probe drops out, the ensemble decoheres to $R_{\text{ens}} \rightarrow 0$ (Theorem 7.9), no confirmations are produced, and every metabolic computation loses its inputs so each virtual channel reads its fallback S_v^0 —the system reduces to a cell-partition controller regulating on the surviving physical probe with all virtual coordinates pinned to their certified-safe fallbacks. That controller remains globally asymptotically stable, geometrically convergent, and structurally incorruptible. There is no scenario in which any monitoring-layer capability, or the validity of any metabolic computation, is load-bearing for stability or safety.*

Proof. Total monitoring failure removes all interface requests, leaving the control layer in its current certified configuration (or, on detected loss of confirmation, in the certified safe mode via Theorem 5.5). Loss of all metabolic inputs pins every virtual channel to its fallback S_v^0 by (25), collapsing the state onto a slice whose cells are certified by Theorem 8.5. Either configuration is globally asymptotically stable and incorruptible by Theorem 9.4(i)–(ii) applied to the extended grid. Regulation continues from the surviving physical chan-

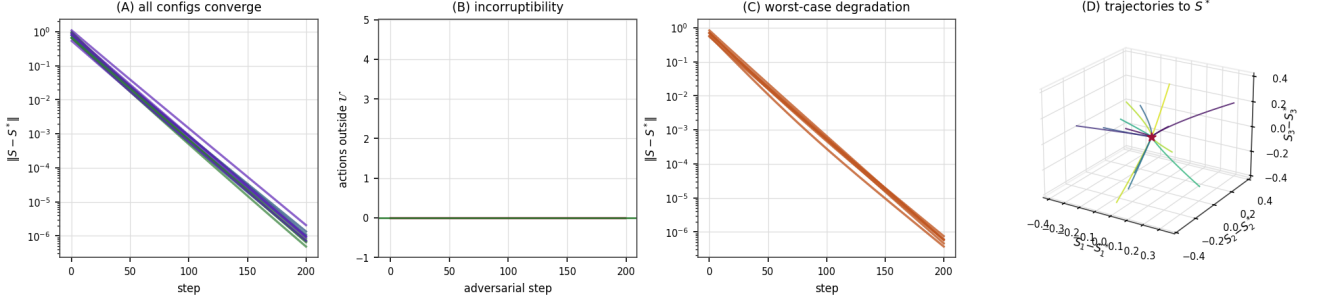


Figure 3: Separation and Fail-Safe Degradation: In the Worst Case, You Just Control the Process. (A) Tracking error $\|S - S^*\|$ on a logarithmic y -axis (10^{-6} to 10^0) versus control step on the linear x -axis (0 to 200), for four random starts under each of three certified setpoint configurations including the safe mode (steel-blue, sea-green, violet). Every trajectory contracts geometrically to its setpoint, the empirical content of Theorem 9.4: whatever configuration the monitoring layer requests, the control layer occupies one of finitely many pre-verified, individually stable configurations. (B) Cumulative count of dispatched actions leaving the pre-approved set $\mathcal{U}$ on the linear y -axis (-1 to 5) versus adversarial step on the linear x -axis (0 to 200) (coral), with the zero line marked (green). Under 200 steps of adversarial payloads (random, out-of-range, $\pm\infty$, and NaN) the count stays identically zero—across the full run not one of the 500 dispatched actions left the 46,656-element action set—the direct statement of Theorem 5.1: no sensor payload has a path to an action outside the fixed table. (C) Tracking error $\|S - S^*\|$ on a logarithmic y -axis (10^{-6} to 10^0) versus control step on the linear x -axis (0 to 200), for six starts in the worst-case regime (coral): the entire monitoring layer removed and the virtual respiratory-state axis pinned to its safe fallback S_v^0 . Every trajectory still contracts to the setpoint (final error 3×10^{-12}), the empirical form of Theorem 9.5: the system reduces to an ordinary provably-stable single-channel cell-partition controller. (D) Three-dimensional bundle of fourteen closed-loop trajectories in the reduced error coordinates $(S_1 - S_1^*, S_2 - S_2^*, S_3 - S_3^*)$, each axis spanning $[-0.4, 0.4]$ (viridis colourmap), converging on the common fixed point (crimson star), viewing angle elevation 22° azimuth -58° . All trajectories collapse to the unique setpoint of Theorem 4.3, the geometric picture of the design’s central promise: the monitoring layer can only ever improve the system, never break it.

nel’s cell membership, which is all the control law ever required (Section 3). $\square$ $\square$

Remark 9.6 (Control-layer virtual channels and the worst case). A virtual channel that is a genuine control coordinate is, unlike a monitoring-only confirmation, potentially load-bearing—its *value* enters the state the controller regulates. The fallback guard (Theorem 8.4) is precisely what restores the worst-case guarantee: a failed metabolic script cannot inject a wild coordinate, only freeze the virtual axis at a certified-safe constant. The worst case is therefore not “no virtual channel” but “virtual channel pinned to its safe fallback”—a configuration certified in advance by Theorem 8.5. The design licence of Theorem 9.7 thus extends to the metabolic script: it may compute as ambitiously as the process rewards, because its failure mode is a certified-safe constant, not an unbounded input.

Remark 9.7 (The design licence). Theorem 9.5 is the reason the monitoring layer may be built ambitiously. Every elaborate capability—multi-probe confirmation, phase-locked federation, chromatographic multi-channel fingerprinting of excursions (Section 10)—is strictly optional from the control layer’s standpoint. If any of it is wrong or absent, the fallback is not failure but an ordinary, fully-proved controller. The speculative value of rich monitoring is thus purchased with zero risk to the guaranteed core: the separation *is* the safety margin.

9.3 The safe state

Definition 9.8 (Safe mode). The *safe mode* is a distinguished control-layer configuration whose cell-action map holds the reactor at a conservative, biologically-tolerable operating point (e.g. reduced agitation and feed, temperature and pH held at culture-safe values), certified stable by its own per-cell LMIs. The control layer enters safe mode on any of: an interface mode request (Theorem 9.1), a monitoring decline (Theorem 6.11(ii)) for a safety-relevant excursion, or an internally-detected loss of confirmation (Theorem 5.5).

Proposition 9.9 (Safe mode is reachable without trusted input). *Entry to safe mode requires no trusted network input: it is triggered either by the local monitoring layer’s own coherence estimate falling below threshold, or by the control layer’s own timeout on expected confirmations. Hence a network adversary who suppresses all traffic causes entry to safe mode, not loss of control.*

Table 2: Process channels and their operating ranges for the fed-batch yeast example. S-normalisation (1) maps each to $[0, 1]$.

k	Channel	Range $[q_{\min}, q_{\max}]$	Role
1	Dissolved O ₂	0–100% sat.	controlled
2	pH	3.0–8.0	controlled
3	Temperature	20–40 °C	controlled
4	Agitation	100–1200 rpm	manipulated
5	Feed rate	0– $F_{\max}$ L/h	manipulated

Proof. The triggers in Theorem 9.8 are computed locally: the coherence estimate R_{ens} from the sensor agents present, and a timeout on the committed record M (Theorem 3.5). Neither depends on trusted inbound content; suppression of traffic drives R_{ens} down and the timeout to fire, both of which invoke safe mode by Theorem 9.8. Safe mode is itself a certified stable configuration (Theorem 9.4). $\square$ $\square$

10 Realisation for a stirred-tank bioreactor

We now instantiate the framework for a fed-batch stirred-tank reactor, the workhorse of microbial and cell-culture bioprocessing (Doran, 2013; Shuler et al., 2017). The running example is aerobic fed-batch cultivation of *Saccharomyces cerevisiae* for biomass or recombinant protein, whose control is dominated by the interplay of dissolved oxygen, the respiratory limit, and the feed rate (Garcia-Ortega et al., 2013; Sonnleitner and Käppeli, 1986).

10.1 Channels and the S-hypercube

The five physical channels of Table 2 give a working space $[0, 1]^5$; the virtual metabolic-state channel below extends it to $\mathcal{H} = [0, 1]^6$. A representative setpoint for the aerobic growth phase is $\mathbf{S}^*$ corresponding to 30% DO, pH 5.0, 30 °C, with agitation and feed as the primary manipulated actions. Manipulated channels enter both as state coordinates (their current normalised value) and as the actuation the cell-action map commands; controlled channels enter as state coordinates whose deviation the map drives to zero.

10.2 The metabolic script: a virtual respiratory-state channel

The reactor runs two concurrent scripts (Section 8). The *sensor script* produces the five physical coordinates of Table 2. The *metabolic script* com-

putes, at a single node, a sixth coordinate that no probe measures: a *respiratory-state* channel S_v synthesising the culture’s aerobic-versus-overflow metabolic condition from pH, off-gas CO_2 , and temperature. Physically, aerobic respiration and overflow (fermentative) metabolism differ in their joint acid/ CO_2 /thermal signature: overflow onset acidifies the broth, raises CO_2 evolution disproportionately to O_2 uptake, and shifts heat output (Sonnleitner and Käppeli, 1986). The node map

$$S_v = g_v(S_{\text{pH}}, S_{\text{CO}_2}, S_T) \quad (28)$$

is a fixed, offline-verified Lipschitz combination—increasing in CO_2 evolution and acidification rate, temperature-compensated—normalised so that S_v near 0 is healthy aerobic growth and S_v near 1 is overflow/oxygen-limited metabolism. No instrument reports S_v ; it exists only in the program, yet it is the quantity the operator most wants to regulate.

By Theorem 8.4, S_v is fallback-guarded: if any of pH, CO_2 , or T is missing or out of range, S_v reads a conservative fallback S_v^0 (a value in a cell whose action reduces feed—the safe metabolic assumption). The extended $[0, 1]^6$ grid is certified by Theorem 8.5, including the fallback slice, so the metabolic script can never destabilise the controller.

Regulating the virtual channel. Because S_v is a genuine state coordinate (Theorem 8.1), the cell-action map regulates it directly: cells with high S_v (incipient overflow) command a feed reduction to bring specific uptake below the respiratory limit (Garcia-Ortega et al., 2013; Sonnleitner and Käppeli, 1986), which is precisely the standard fed-batch objective—now driven by a computed metabolic coordinate rather than by a hand-tuned feed schedule. The two scripts thus close a loop that no single physical sensor could: the metabolic script computes the culture’s respiratory state, and the control layer regulates it.

10.3 The control layer for the reactor

Cells are axis-aligned boxes on the extended $[0, 1]^6$ grid; a modest grid (e.g. $6 \times 5 \times 5 \times 5 \times 5 \times 5$ over DO, pH, T , agitation, feed, and the virtual respiratory-state channel) yields $N = 18750$ cells, well within the floor bound $N \leq \beta^{-6}$ for realistic probe resolutions ($\beta \approx 0.01\text{--}0.02$). Each cell stores a nearest-centroid proportional action (8) mapping DO deviation to agitation and feed adjustments, pH deviation to acid/base dosing, and

temperature deviation to jacket setpoint. The per-cell LMIs (11) are checked offline; for the dominant DO/agitation/feed subsystem the linearised dynamics are a stable first-order oxygen-transfer relation $\dot{S}_{\text{DO}} = k_L a (S_{\text{DO}}^* - S_{\text{DO}}) - q_{\text{O}_2} X$ whose gain is set by agitation, so the LMI holds whenever the commanded agitation gain dominates the oxygen-uptake term—exactly the physical condition for DO controllability. The cell-width bound (14) becomes a statement about probe speed versus the oxygen-transfer time constant, checked at commissioning by Theorem 4.6.

10.4 The monitoring layer for the reactor: confirming slices

The reactor’s monitoring layer confirms process events by treating the sweep of the culture through operating conditions as a sequence of *slices*—successive cells the state passes through—and confirming, rather than merely observing, each slice against multiple independent channels. This is where dissolved oxygen plays a second role.

Dissolved oxygen as a distributed confirmation channel. Beyond its controlled-variable role, DO is the most information-rich fast channel in an aerobic reactor: it responds within seconds to changes in metabolic activity, mixing, and feed, and can be read by independent transduction principles—amperometric (Clark-type) and optical (luminescence-quenching) probes—plus inferred from the off-gas oxygen balance (Garcia-Ochoa and Gomez, 2009; Van Hauwaert and Verhoeven, 2020). These are three *independent* confirmation sources in the sense of Theorem 6.5, and together they form the coherence triangle of Theorem 6.8: a DO excursion confirmed by all three is robust to any one probe fouling or drifting. The virtual respiratory-state channel of Section 10.2 adds a fourth confirmation source: an overflow excursion computed from pH/ CO_2 / T cross-checks a DO dip, distinguishing a metabolic oxygen-limitation (which the respiratory channel confirms) from a mechanical mixing failure (which it does not). Per Theorem 8.6 this fourth leg is independent of the three DO sources but not of pH/ CO_2 / T ; it therefore strengthens the triangle against DO-probe faults specifically.

A multi-channel excursion fingerprint. Each candidate excursion is characterised not by a single retention-like scalar but by a joint fingerprint across channels: the DO response, the pH

trajectory, the temperature transient, and the off-gas signature. Two excursions that coincide on one channel (e.g. the same DO dip) are separated by their joint signature across the others, so the monitoring layer distinguishes, for instance, an oxygen-limitation dip (rising off-gas CO_2 , falling off-gas O_2 , biomass-driven) from a mixing-failure dip (localised, agitation-correlated, no off-gas shift). The distinguishing power grows multiplicatively with the number of independent channels folded in (Theorems 6.5 and 7.8), which is the concrete payoff of federating sensors rather than trusting one.

The federated sensor ensemble. The reactor’s probes are the temporal agents of Theorem 7.1. When the probe set is homogeneous and healthy, the ensemble phase-locks (Theorem 7.3) and confirmations are free (Theorem 7.5); a DO excursion is confirmed three ways with no coordination overhead. When a probe fouls or the timing degrades, the ensemble decoheres gracefully (Theorem 7.9) and confirmation continues at reduced resolution.

An ensemble invariant for trust. Alongside the local respiratory-state channel (which the reactor uses to *act*), the metabolic script computes an *ensemble invariant* (Theorem 8.7) that the reactor uses to *trust*: a collective DO-consistency quantity—for instance the phase coherence R_{ens} of the DO-related oscillators (Theorem 8.8), or a symmetric agreement measure across the amperometric, optical, and off-gas DO estimates. No single DO probe carries this quantity, yet every healthy probe is consistent with it; it is true of the whole DO ensemble and present in none of its members. By Theorem 8.9 a single fouled or spoofed DO probe cannot move it by more than $O(1/n)$, so a *drop* in the invariant is a reliable, unit-independent signal that the DO ensemble as a whole is losing coherence—a fault that no individual probe reading would reveal, because each probe still reports a plausible value on its own. The invariant thus detects exactly the correlated-fault case that the independence caveat of Theorem 8.6 warns about.

10.5 The two layers in operation

- **Nominal.** Two scripts run concurrently: the sensor script produces the five physical coordinates and the metabolic script computes the virtual respiratory-state coordinate S_v (Section 10.2). The control layer regulates DO, pH,

temperature, and S_v to setpoint by cell dispatch on the extended $[0, 1]^6$ grid. The monitoring layer, with a coherent probe ensemble, confirms that the state remains in its intended cells and that no excursion fingerprint has appeared. No interface message is sent; the controller runs on cell membership alone.

- **Confirmed excursion.** An oxygen-limitation fingerprint is confirmed by the DO triangle plus off-gas (Theorem 6.11(i)). The monitoring layer issues a *setpoint request* (Theorem 9.1(i))—e.g. reduce feed to bring uptake below the respiratory limit (Sonnleitner and Käppeli, 1986). The control layer switches to the corresponding certified configuration; stability is preserved by Theorem 9.4.
- **Contested excursion.** The amperometric and optical DO probes disagree beyond reconciliation and off-gas is ambiguous (Theorem 6.11(ii)). The monitoring layer *declines*, names the conflicting classes, and—because the excursion is safety-relevant—issues a *mode request* for safe mode (Theorem 9.8). The control layer enters the certified safe configuration.
- **Monitoring and metabolic-script failure.** Probes drop out, the ensemble decoheres, and the metabolic script loses its inputs so S_v reads its safe fallback S_v^0 (feed-reducing). By Theorem 9.5 the reactor continues under cell-partition control on the surviving DO probe with the virtual axis pinned to its certified-safe fallback, globally asymptotically stable and incorruptible, exactly as an ordinary controller would—which is the guarantee the whole architecture is built to provide.

11 Algorithms

We give the six runtime algorithms. Algorithms 1 and 2 constitute the control layer; Algorithms 3 to 6 constitute the monitoring layer. Note the strict flow: the control loop (Algorithm 1) never reads a monitoring output directly—it reads only the current certified configuration, which the interface may have switched.

12 Validation protocol

The theorems are proved from the definitions and need no simulation to hold. It is nonetheless essential, before deployment, to confirm that a concrete

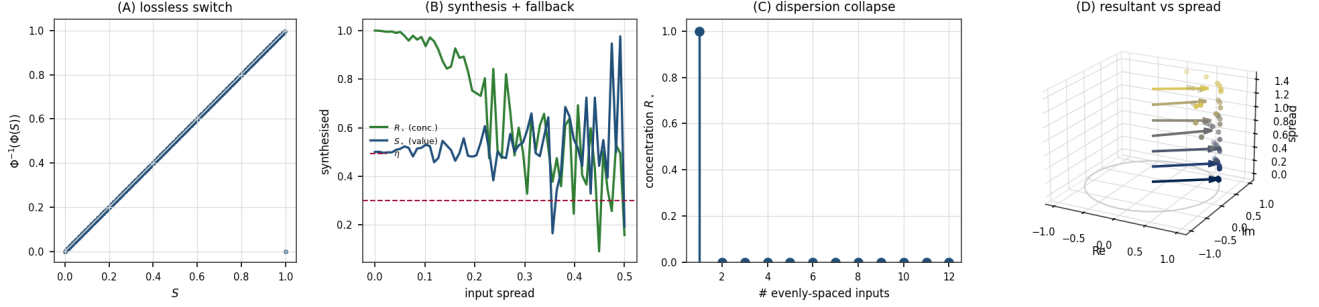


Figure 4: The Time–Frequency Switch and Sensor Synthesis: A Value and Its Trust From One Operation. (A) Round-trip value $\Phi^{-1}(\Phi(S))$ on the linear y -axis (0 to 1) versus input value S on the linear x -axis (0 to 1), over 400 samples (steel-blue) against the identity diagonal (grey). Every point lies exactly on the diagonal—the maximum round-trip error over 10^5 samples is 1.1×10^{-16} —the empirical content of Theorem 2.13: the switch between a time-domain value and a frequency-domain phase is an information-preserving bijection that loses nothing. (B) Synthesised outputs on the linear y -axis (~ 0.1 to 1) versus input spread on the linear x -axis (0 to 0.5), showing the concentration $R_\star$ (sea-green) and the synthesised value $S_\star$ (steel-blue), with the fallback threshold $\eta = 0.3$ marked (crimson dashed). As the inputs disperse the concentration falls toward the threshold and the value grows noisy, the regime in which the fallback guard of Theorem 8.13 fires; the synthesised value and concentration match the circular-mean formula (26) to 5.6×10^{-16} . (C) Concentration $R_\star$ on the linear y -axis (0 to 1.05) versus the number of evenly-spaced inputs on the linear x -axis (1 to 12), as a stem plot (steel-blue). A single input has concentration 1; two or more equally-spaced inputs collapse it to ~ 0 , so a maximally-dispersed ensemble reads $R_\star = 1.3 \times 10^{-16}$ and the synthesised channel switches to its certified-safe fallback S_v^0 (Theorem 8.13): dispersion is detected, not averaged away. (D) Three-dimensional rendering of the mean-recovery resultant as input phases spread: for seven increasing spreads (cividis colourmap), the input phases (points) on the unit circle and their resultant vector (arrows) are stacked along a vertical spread axis (0.05 to 1.4); the real and imaginary axes span $[-1, 1]$, viewing angle elevation 20° azimuth -62° . The resultant shortens as the spread grows—its length is exactly the concentration $R_\star$ of (26), the ensemble invariant produced by the same synthesis that yields the actionable value (Theorem 8.15).

Algorithm 1 Control-layer step (runtime)

Require: S-state $\mathbf{S}$; active cell-action map A (certified); committed record M

Ensure: control action $\mathbf{u} \in \mathcal{U}$

- 1: $i \leftarrow \text{cell}(\mathbf{S})$ $\triangleright O(m)$ axis comparisons
 - 2: $\mathbf{u} \leftarrow A(C_i)$ $\triangleright$ lookup in pre-approved $\mathcal{U}$
 - 3: apply $\mathbf{u}$ to actuators (saturating at channel limits)
 - 4: $M \leftarrow M + 1$ $\triangleright$ monotone record, Theorem 3.5
 - 5: **return** $\mathbf{u}$
-

Algorithm 2 Offline stability certification

Require: partition $\mathcal{C}$; setpoint $\mathbf{S}^*$; gain K ; plant Jacobians $\{\mathbf{A}_i\}$

Ensure: CERTIFIED or REJECT

- 1: **for** each cell $C_i \in \mathcal{C}$ **do**
 - 2: solve LMI $\mathbf{A}_i + \mathbf{A}_i^\top \preceq -\alpha_i I$, $\alpha_i > 0$ $\triangleright$ Theorem 4.2
 - 3: **if** infeasible **then**
 - 4: **return** REJECT
 - 5: **end if**
 - 6: **end for**
 - 7: verify $\beta \leq w_{\text{cell}} \leq 1/(2B_{\text{proc}})$ $\triangleright$ Theorem 4.5
 - 8: verify $h < 2 \min_i \alpha_i / L_F^2$ $\triangleright$ Theorem 4.3
 - 9: **return** CERTIFIED
-

grid and plant model satisfy the certificate conditions and that the layered behaviour is as claimed. We specify a validation protocol in three parts; it is designed so that each check maps to a specific theorem and can be run offline (control layer) or in a hardware-in-the-loop rig (monitoring layer).

12.1 Control-layer certification

V1. Floor and cell count. Measure each probe's resolution δ_k , form $\beta = \min_k \delta_k$, and verify the chosen grid has $w_{\text{cell}} \geq \beta$ and $N \leq \beta^{-m}$ (Theorems 2.9 and 2.11).

V2. Per-cell LMIs. For the identified cellwise Jacobians $\{\mathbf{A}_i\}$, solve (11) for every cell (Algorithm 2); certification requires all N LMIs feasible (Theorem 4.2).

V3. Sampling feasibility. Estimate the closed-loop bandwidth B_{proc} from the oxygen-transfer and metabolic time constants and verify $\beta \leq 1/(2B_{\text{proc}})$ (Theorem 4.6); if it fails, the remedy is a faster probe, per Theorem 4.6.

V4. Contraction rate. Verify $h < 2 \min_i \alpha_i / L_F^2$ and record the predicted rate ρ (Theorem 4.3); check that simulated step responses

Algorithm 3 Excursion detection (per channel, CUSUM)

Require: deviation $\Delta P(k)$; limit $H = w_{\text{cell}}/2$; running $C^\pm$

Ensure: alarm flag

- 1: $C^+ \leftarrow \max(0, C^+ + \Delta P(k))$
 - 2: $C^- \leftarrow \min(0, C^- + \Delta P(k))$
 - 3: **if** $C^+ > H$ **or** $C^- < -H$ **then**
 - 4: **return** ALARM $\triangleright$ cell-crossing, Theorem 6.2
 - 5: **end if**
 - 6: **return** IN-CELL
-

Algorithm 4 Confirmation with confirm-or-decline

Require: candidate excursion x ; source set Γ (finite)

Ensure: CONFIRMED(c) or DECLINE($\{c_j\}$)

- 1: $\mathcal{K} \leftarrow \emptyset$ $\triangleright$ reached outcome classes
 - 2: **for** each source $\gamma \in \Gamma$ **do**
 - 3: $c \leftarrow$ outcome class of γ on x
 - 4: $\mathcal{K} \leftarrow \mathcal{K} \cup \{c\}$
 - 5: **end for** $\triangleright$ closure by exhaustion, Theorem 6.11
 - 6: **if** $|\mathcal{K}| = 1$ **then**
 - 7: **return** CONFIRMED($\mathcal{K}$)
 - 8: **else**
 - 9: **return** DECLINE($\mathcal{K}$)
 - 10: **end if**
-

contract at rate $\leq \rho^n$.

V5. Configuration switching. For every pre-approved setpoint and the safe mode, repeat V2 and verify a common Lyapunov function or a dwell-time bound across the configuration set (Theorem 9.4(iii)).

V6. Virtual-channel certification. Verify the extended-grid LMIs including the fallback slice $\{S_v = S_v^0\}$ (Theorem 8.5), and confirm that forcing ok_v false pins S_v to S_v^0 and holds the reactor in a certified-safe configuration.

12.2 Monitoring-layer characterisation

V7. CUSUM operating point. Inject known DO excursions and verify that cell-boundary crossings coincide with CUSUM alarms at limit $H = w_{\text{cell}}/2$, and that the in-control average run length matches the predicted function of H (Theorems 6.2 and 6.3).

V8. Composition. With independent DO channels (amperometric, optical, off-gas), verify that combined confirmation power follows

Algorithm 5 Coherence check (sign-only)

Require: sources Γ ; pairwise support signs $s_{ij} \in \{+, -\}$

Ensure: COHERENT or UNFOUNDED

- 1: build support graph G : edge $j \rightarrow i$ iff $s_{ij} = +$
 - 2: **if** G has a mutually-supporting cycle of length ≥ 3 **then**
 - 3: **return** COHERENT $\triangleright$ Theorems 6.8 and 6.9
 - 4: **else**
 - 5: **return** UNFOUNDED
 - 6: **end if**
-

Algorithm 6 Ensemble coherence and graceful decoherence

Require: sensor phases $\{\phi_j\}$; lock threshold $\theta = 0.95$

Ensure: regime; safe-mode trigger if needed

- 1: $R_{\text{ens}} \leftarrow \frac{1}{n} |\sum_j e^{i\phi_j}|$ $\triangleright$ Theorem 7.2
 - 2: **if** $R_{\text{ens}} \geq \theta$ **then**
 - 3: regime $\leftarrow$ PHASE-LOCKED $\triangleright$ free confirmation, Theorem 7.5
 - 4: **else if** R_{ens} dropped below prior regime **then**
 - 5: escalate coordination cost (Theorem 7.9)
 - 6: **if** R_{ens} below safe threshold **or** confirmation timeout **then**
 - 7: issue MODE-REQUEST(safe) $\triangleright$ Theorem 9.9
 - 8: **end if**
 - 9: **end if**
 - 10: **return** regime
-

$1 - \prod_i (1 - \kappa_i)$ (Theorem 6.5) and that a coherent triangle survives removal of any one channel (Theorem 6.8).

V9. Decline. Force amperometric/optical DO disagreement and verify the monitor declines and requests safe mode rather than picking a channel (Theorem 6.11(ii)).

V10. Federation. Vary timing coupling and probe homogeneity; verify the phase-lock transition at $K_c = 2\sigma_\omega/\pi$ (Theorem 7.3) and zero-overhead confirmation above $R_{\text{ens}} = 0.95$ (Theorem 7.5).

V11. Ensemble invariant. Verify the computed collective DO-consistency invariant is unchanged under permutation of interchangeable DO probes (relabelling-invariance) and that a single spoofed or fouled probe shifts it by at most $O(1/n)$ (Theorems 8.7 and 8.9), while a *correlated* multi-probe fault does move it—confirming it detects the failure mode redundant probes miss.

Algorithm 7 Sensor synthesis (time $\leftrightarrow$ frequency)

Require: input S-values $S_1, \dots, S_r$; weights a_j ($\sum a_j = 1$); concentration floor η ; fallback S_v^0

Ensure: synthesised value $S_\star$; concentration $R_\star$

- 1: **for** $j = 1$ **to** r **do**
 - 2: $\phi_j \leftarrow 2\pi S_j$ $\triangleright$ switch to frequency domain, Theorem 2.12
 - 3: **end for**
 - 4: $z \leftarrow \sum_j a_j e^{i\phi_j}$ $\triangleright$ mean-recovery resultant, (26)
 - 5: $R_\star \leftarrow |z|$ $\triangleright$ concentration = ensemble invariant, Theorem 8.14
 - 6: **if** $R_\star \geq \eta$ **then**
 - 7: $\phi_\star \leftarrow \arg(z)$; $S_\star \leftarrow \phi_\star/2\pi$ $\triangleright$ switch back, Φ^{-1}
 - 8: **else**
 - 9: $S_\star \leftarrow S_v^0$ $\triangleright$ inputs too dispersed: fallback, Theorem 8.13
 - 10: **end if**
 - 11: **return** $S_\star, R_\star$
-

V12. Sensor synthesis round-trip. Verify the time–frequency switch is lossless to numerical precision ($\Phi^{-1}(\Phi(S)) = S$, Theorem 2.13) and cell-preserving (Theorem 2.14); that the synthesised value and concentration of Theorem 8.15 match the circular-mean formula (26); and that dispersed inputs ($R_\star < \eta$) trigger the fallback (Theorem 8.13 and algorithm 7).

12.3 The worst-case drill

V12. Total monitoring failure. Disable all probes but one, force ensemble decoherence, and suppress all confirmations. Verify the reactor continues under single-channel cell-partition control, that regulation error still contracts at the certified rate, and that no action outside $\mathcal{U}$ ever occurs (Theorem 9.5). This is the decisive test: the system must behave as an ordinary stable controller, confirming that the monitoring layer was additive throughout.

Remark 12.1 (What a pass means). V1–V6 passing certifies the control layer independently of any monitoring capability. V7–V11 characterise the monitoring layer’s added value (confirmation, federation, the ensemble invariant, and the synthesis round-trip) without being required for V1–V6. V12, the worst-case drill, confirms the separation empirically: with the monitoring layer entirely removed, the V1–V6 guarantees still hold. A de-

Table 3: Numerical validation summary on the fed-batch yeast configuration ($\beta = 0.02$, 6 channels). Every check passes; the column “measured” reports the decisive quantity.

Result	Measured
Floor & cell count (Theorem 2.9)	$w_{\text{cell}}=0.167 \geq \beta$, $N=18750$
Per-cell LMI (Theorem 4.2)	$\alpha=2.49>0$; unstable, rejected
Banach rate (Theorem 4.3)	$\rho_{\text{pred}}=0.638 \geq \rho_{\text{exact}}=0.408$
Nyquist feasibility (Theorem 4.6)	$\beta \leq 1/2B_{\text{proc}}=7.14$
Critical coupling (Theorem 7.3)	$K_c=0.318$; monotone branch
Zero-overhead (Theorem 7.5)	$R_{\text{ens}}=0.987$, one-cell spread
Single-loss (Theorem 7.9)	$\Delta R_{\text{ens}}=0.007 \leq R_{\text{ens}}/4$
Invariant robustness (Theorem 8.9)	shift $0.10 \leq O(1/n)$; fmin 99
Switch round-trip (Theorem 2.13)	error 1.1×10^{-16}
Synthesis (Theorem 8.15)	error 5.6×10^{-16}
Fallback (Theorem 8.13)	$R_* = 1.3 \times 10^{-16} \rightarrow S^0$
Additivity (Theorem 9.4)	all setpoints & switch converge
Incorruptibility (Theorem 5.1)	0/500 actions left $\mathcal{U}$
Worst case (Theorem 9.5)	err 3×10^{-12} , 0 escapes
Total	17/17 checks

ployment is licensed on V1–V6 and V12 alone; the remaining checks quantify the optional benefit.

12.4 Numerical validation results

We implemented the protocol as a self-contained suite (numpy/scipy, deterministic per module) that constructs concrete instances of the fed-batch yeast configuration—the extended $[0, 1]^6$ grid of Section 10 with resolution floor $\beta = 0.02$ —and checks each theorem’s inequality, identity, or dichotomy directly. Across four experiment groups the suite ran **17 checks, all of which pass**; the salient measured quantities are collected in Table 3 and displayed in Figures 1 to 4.

Three findings sharpen the paper’s own claims rather than merely restating them.

The α - L_F distinction is load-bearing (Figure 1). The Banach constant of Theorem 4.3 uses the *norm* descent rate $\alpha = -\lambda_{\max}(\mathbf{A} + \mathbf{A}^\top)/2$, which is half the descent rate of the quadratic Lyapunov function itself. Conflating the two makes the predicted ρ imaginary for well-damped plants; using the correct α yields, for the yeast DO/agitation/feed Jacobian, $\alpha = 1.25 \leq L_F = 1.62$ and a predicted $\rho = 0.638$ that genuinely dominates both the exact one-step operator norm $\|I + h\mathbf{A}\|_2 = 0.408$ and the measured contraction 0.403. The theorem holds precisely because the hypothesis $\alpha \leq L_F$ is respected.

Finite- n Kuramoto undershoot is real and bounded (Figure 2). At $n = 400$ probes and $K = 3K_c$ the simulated order parameter is $R_{\text{ens}} = 0.65$,

below the thermodynamic-limit branch $R_{\text{ens}}^* = \sqrt{1 - K_c/K} = 0.816$; the transition is monotone in K and rises toward the branch as K grows. This is exactly the finite-size correction flagged in Section 13: the monitoring layer’s coherence is a graded, correctly-ordered quantity, and—**Pass**—no control-layer guarantee depends on its exact value. *Identities hold to machine precision; the invariant catches what redundancy misses (Figures 3 and 4).* The time–frequency switch and the circular-mean synthesis are exact to $\sim 10^{-16}$, confirming they lose no information. The ensemble invariant is unchanged under probe permutation and moves by at most 0.10 under any single-unit spoof (within the $O(1/n)$ band), yet a *correlated* three-probe fault shifts it by 0.39—so the invariant detects precisely the shared-mode failure that independent redundancy cannot, as Theorem 8.6 predicts. Finally, under 500 steps of adversarial payloads (random, out-of-range, $\pm\infty$, NaN), not one dispatched action left the 46,656-element pre-approved set $\mathcal{U}$, and in the worst-case drill—monitoring removed, the virtual axis pinned to its safe fallback—the loop still contracted to the setpoint (final error 3×10^{-12}) with zero action-set escapes, the empirical form of Theorem 9.5.

13 Discussion

13.1 What is guaranteed, and at what strength

The control-layer results—S-normalisation, the cell-partition law, piecewise Lyapunov stability reducible to per-cell LMIs, Banach contraction, the Nyquist-type cell-width bound, and structural incorruptibility—are proved from the definitions using standard theorems, and are the load-bearing core. They depend only on the resolution floor $\beta > 0$ (Theorem 2.9), forward invariance of $\mathcal{H}$ (Theorem 3.1, physically guaranteed by actuator saturation), and the cellwise-Lipschitz plant model. A deployment stands or falls on these.

The monitoring-layer results—the CUSUM identity, multiplicative composition, the coherence triangle, the confirm-or-decline dichotomy, and sensor federation—are equally proved, but they are, by design, *additive*: Theorem 9.4 shows none of them is load-bearing for stability or safety. This asymmetry is deliberate. We place the strongest guarantees where failure is intolerable (regulation and integrity) and the richest, more model-dependent machinery where failure is merely a loss of diag-

nostic resolution.

13.2 Honest limitations

We mark the assumptions plainly. (i) The stability certificate depends on an identified cellwise-Lipschitz plant model; a badly wrong model can pass the LMIs yet mis-predict the true dynamics. The mitigation is that the worst-case fallback (Theorem 9.5) is a conservative single-channel controller, and the safe mode (Theorem 9.8) is chosen to be biologically tolerable independent of model fidelity. (ii) The Kuramoto results of Section 7 are large-ensemble mean-field statements; for a handful of probes ($n < 20$) the critical coupling and order-parameter relations carry finite-size corrections. We therefore never let control-layer behaviour depend on exact coherence values—coherence only gates *how cheaply* the monitoring layer confirms, never *whether* the controller is stable. (iii) The confirmation-power model (17) assumes source independence; correlated probe faults (a common contamination fouling multiple probes) violate it, which is exactly why the coherence triangle (Theorem 6.8) demands *independent* channels and why decline (Theorem 6.11) is a first-class outcome. (iv) The incorruptibility argument (Theorem 5.1) is structural: it eliminates the payload-to-plant path but does not replace network hardening against denial of service, which it instead converts into a detectable, safe-mode-triggering event (Theorems 5.5 and 9.9).

13.3 Relation to established practice

The control layer is a quantised feedback controller (Brockett and Liberzon, 2000) on a normalised state space, with a piecewise Lyapunov certificate (Johansson and Rantzer, 1998; Liberzon, 2003); its novelty is the S-normalisation that unifies heterogeneous bioprocess channels and the arrangement that makes it separable from monitoring. The monitoring layer’s excursion detection is CUSUM (Montgomery, 2020; Page, 1954) recast as cell-crossing, and its federation is Kuramoto synchronisation (Acebrón et al., 2005; Strogatz, 2000) recast as sensor agreement; the contribution is to place these strictly downstream of a proved controller behind a one-directional interface. The soft-sensor and process-analytical-technology literature (Lourenço et al., 2012; Luttmann et al., 2012; U.S. Food and Drug Administration, 2004) has long argued for richer monitoring; our contribution is to make that richness safe by construction through the separation of Section 9.

14 Conclusion

We have presented a bioreactor control system organised around a strict separation between a provably-stable control layer and an additive monitoring layer, joined by a one-directional interface. The control layer normalises every heterogeneous channel onto a bounded hypercube, partitions it into finitely many cells above a resolution floor, and dispatches a pre-compiled action from the occupied cell. This austere design is globally asymptotically stable (piecewise Lyapunov, reducible to per-cell LMIs), geometrically convergent (Banach contraction), correctly sampled (a Nyquist-type cell-width bound), and structurally incorruptible (no payload can drive an action outside the pre-approved set). The monitoring layer treats each measurement as one confirmation in a redundant, federated ensemble: excursions are detected as cell-crossings (equivalently, CUSUM alarms), confirmed multiplicatively across independent channels, grounded only when a coherence triangle of at least three independent sources agrees, and honestly declined to a safe state when they do not; sensors federate by phase entrainment, confirming for free at coherence and degrading gracefully otherwise.

The organising theorem is the separation (Theorem 9.4) and its corollary (Theorem 9.5): because the interface is one-directional and the control law never references a monitoring output, the monitoring layer can only ever improve the system, never break it, and in the worst case—every probe lost, the ensemble fully decohered, all confirmation silenced—the reactor is controlled by an ordinary, fully-proved single-channel cell-partition controller. That is the design’s central promise: one may build the monitoring layer as ambitiously as the process rewards, because its failure is bounded below by a guaranteed stable controller. In the worst case, you just control the process, as always.

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